Imperial College London
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ROBOTICS: 3D Kinematics lecture notes

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List of Figures

Chapter 1

Frame rotation and translation

1.1 Resources

Watch this video from 0:00-11:13 seconds.

1.2 Why does a design engineer need to know about this?

Figure 1.1: Different types of robots sharing the same 3D kinematics problem

Design engineers have to work with 3D articulated structures not only in robotics, but also in other mechanical and animated systems to deliver services. These can range from collapsible utensils, furniture, to automobile parts and modern architecture. In this module we focus on applications in articulated robots used to reach in 3D space to deliver useful services in industrial settings. Irrespective of the specific type of robot (see figure 1.1 for some examples), they all face the same problem of having to know where different articulated limbs are at any given time, so that the robot can coordinate their movements to provide a meaningful service. In this lecture we study some mathematical tools we can use to solve this problem.

1.3 Intended learning outcomes

At the end of this lecture you should be able to

1. Know the structure and properties of a 3D rotation matrix Know what a homogeneous coordinate system is
2. Know the structure of a homogeneous transformation matrix Know what is meant by linearly independent axes are
3. Know the physical meaning of an Euler rotation
4. Know the difference between successive Euler rotations around moving axes (Euler angles) and fixed axes (Roll, Yaw, and Pitch)
5. Know what a unit quaternion is and how to perform 3D rotations with quaternions
6. Know the properties of homogeneous transformation matrices and their products
7. Know how to perform a compound transformation (a series of 3D transformations)
8. Know how to obtain the Jacobian matrix of a manipulator from the compound homogeneous transformation matrix

1.4 Understanding frames

Figure 1.2 shows how joint encoders are usually mounted at each joint of a robot. The important thing to remember is that each joint only knows how much the link attached to it rotates.

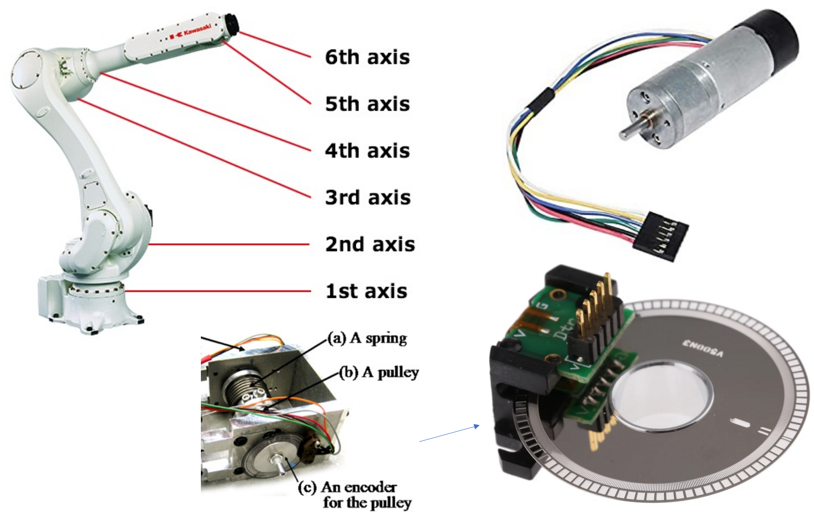


Figure 1.2: Joint encoders

When you look at some object in the room, you see things relative to a coordinate system fixed in your eyes. Then to relate that object to the body and where you are, you have to

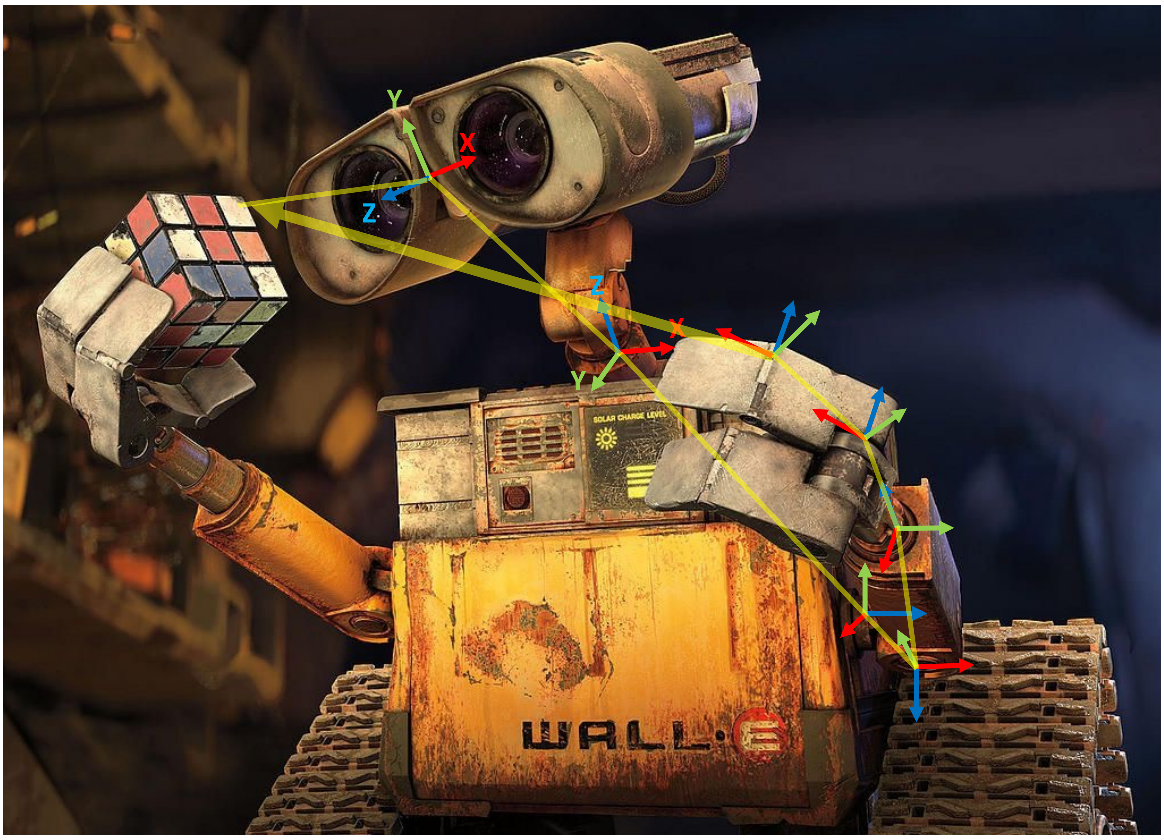


Figure 1.3: In a robot, each joint has sensors to see what is immediately before them relative to the sensor based coordinate frame. When sensors are mounted on a physical body, the each sensor should interpret what the sensor before it relays after transforming the coordinate systems to a common frame. Usually, ith sensor interprets the data of the $(i-1)$th sensor, by transforming the coordinate frame of $(i-1)$th sensor to that of i.

transform the eye coordinate system to the head coordinate system, then to the body coordinate system, and finally to the space coordinate system of the room based on sensors in the muscles (spindle sensors) that function as joint encoders.

Imagine what Wall-e shown in figure 1.3 has to do to relate what the eyes see to its fingers to plan a reaching movement. The challenge it has is that the coordinate frames attached to each joint move and rotate with the posture of the body. Therefore, these coordinate transformation calculations have to be done in each sampling step. This is why it is difficult to execute realtime action in robots in natural environments.

A frame

A coordinate frame i consists of an origin, denoted O_i, and three mutually orthogonal basis vectors, denoted (x_i, y_i, z_i), that are fixed within a given body. In figure 1.4, the origin o_3 of frame (x_3, y_3, z_3) is located at vector p from the frame (x_1, y_1, z_1) with origin o_1. Therefore, frames give a mathematical basis to compute distance and orientation between two frames.

The forward kinematics problem

Given knowledge of only the joint angles and link lengths, the problem of determining the position and orientation of the end-effector of a robot relative the the base is known as the

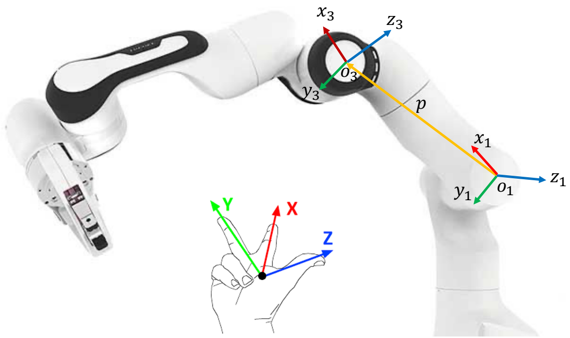


Figure 1.4: Frame rotation around $\hat{z}$ axis

forward kinematic problem. Figure 1.5 shows an example for a Franka robot. A camera mounted at the gripper of the robot sees a cylindrical object to be grasped. However, the gripper cannot grasp it unless all the other joints rotate to position the gripper at the right orientation to grasp it. How can the other joints accomplish this?

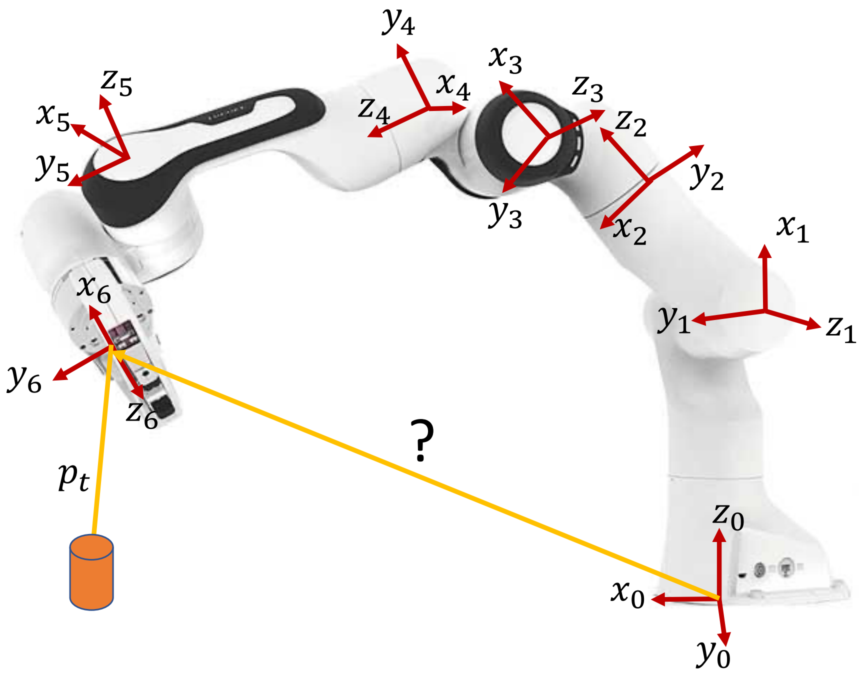


Figure 1.5: Forward kinematics problem in a robot manipulator

The robot can do this by knowing the position and orientation of the object and those of the gripper in the same base frame given by (x_0, y_0, z_0). Once this frame transformation is done, position and orientation errors can be computed at any given time to generate control commands to move the robot in the right direction.

1.4.1 Translation and rotation in 2D

Pure translation in 2D

First it is important to know what is meant by a homogeneous coordinate system. Vector $\begin{bmatrix} X_B \\ Y_B \end{bmatrix}$ is in Euclidean coordinates with a defined origin.

A vector $OP = \begin{bmatrix} kX_B \\ kY_B \\ k \end{bmatrix}$, is known to be in **homogeneous** coordinates, because the origin depends on k. When $k = 1$, $OP = \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix}$.

A pure translation in 2D is shown in figure 1.6.

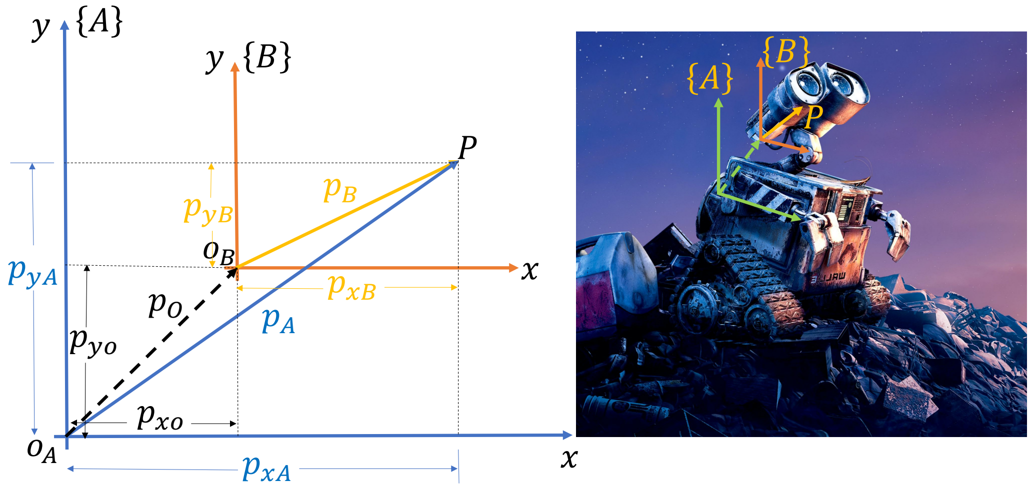


Figure 1.6: Pure translation in 2D

$$\boldsymbol{p}_A = \boldsymbol{p}_0 + \boldsymbol{p}_B \tag{1.1}$$

$$\boldsymbol{p}_A = \begin{bmatrix} p_{xA} \\ p_{yA} \end{bmatrix}, \quad \boldsymbol{p}_O = \begin{bmatrix} p_{x0} \\ p_{y0} \end{bmatrix}, \quad \boldsymbol{p}_B = \begin{bmatrix} p_{xB} \\ p_{yB} \end{bmatrix} \tag{1.2}$$

We can write equation 1.2 in homogeneous coordinates given by

$$\begin{aligned}
\begin{bmatrix} p_{xA} \\ p_{yA} \\ 1 \end{bmatrix} &= \begin{bmatrix} p_{x0} \\ p_{y0} \\ 1 \end{bmatrix} + \begin{bmatrix} p_{xB} \\ p_{yB} \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} p_{x0} + p_{xB} \\ p_{y0} + p_{yB} \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \times p_{xB} + 0 \times p_{yB} + 1 \times p_{x0} \\ 0 \times p_{xB} + 1 \times p_{yB} + 1 \times p_{y0} \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & p_{x0} \\ 0 & 1 & p_{y0} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_{xB} \\ p_{yB} \\ 1 \end{bmatrix} \qquad (1.3)
\end{aligned}$$

The matrix ${}^A\boldsymbol{T}_B = \begin{bmatrix} 1 & 0 & p_{x0} \\ 0 & 1 & p_{y0} \\ 0 & 0 & 1 \end{bmatrix}$ in equation 1.3 is known as the **homogeneous transformation matrix** that maps the vector $\begin{bmatrix} p_{xB} \\ p_{yB} \\ 1 \end{bmatrix}$ of point P in frame $\{B\}$ to a vector in frame $\{A\}$ written in **homogeneous coordinates**.

At this point, it is worth noticing the structure of the matrix ${}^{A}\boldsymbol{T}_{B}$. Notice that coordinates of $\boldsymbol{p}_o$ are in the last column of the transformation matrix. The identity matrix in the 1st and 2nd columns represents the relative rotation between the two coordinate frames. It is an identity matrix in a pure translation.

Geometric interpretation of rotation matrix in 2D

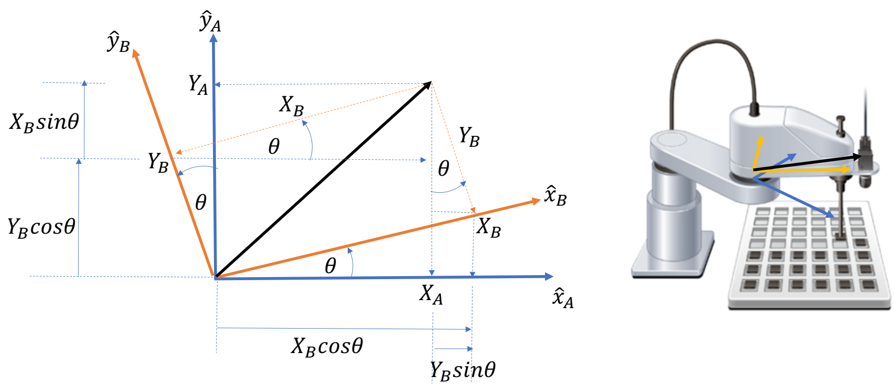


Figure 1.7: The geometric interpretation of the rotation using vector projections.

Figure 1.7 shows how we can make sense of the rotation matrix using geometric projections of vectors from one coordinate system to the other.

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} X_B \cos\theta - Y_B \sin\theta \\ X_B \sin\theta + Y_B \cos\theta \\ 1 \end{bmatrix} \tag{1.4}$$

Then, we can re-arrange the terms so that we get coordinates in frame $\{B\}$ multiplied by a transformation matrix on the right hand, and the coordinates in frame $\{A\}$ on the left hand side of the equation.

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \tag{1.5}$$

We can convert all sin terms in equation 1.5 to cos terms to see a pattern on dot products between the axes being rotated.

Equation 1.5 in cos terms is given by

$$\begin{aligned} \begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} &= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta & \cos(90+\theta) & 0 \\ \cos(90-\theta) & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \end{aligned} \tag{1.6}$$

We can notice from equation 1.6 that the rotation matrix can also be obtained by dot products of unit vectors along each axis:

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \hat{x}_B \cdot \hat{x}_A & \hat{y}_B \cdot \hat{x}_A & 0 \\ \hat{x}_B \cdot \hat{y}_A & \hat{y}_B \cdot \hat{y}_A & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \tag{1.7}$$

where, $\hat{x}_A$, $\hat{x}_B$, $\hat{y}_A$, and $\hat{y}_B$ are unit vectors (magnitude = 1). This is because the angle between $\hat{x}_B$ and $\hat{x}_A$ is θ, that between $\hat{y}_B$ and $\hat{x}_A$ is $(90+\theta)$, that between $\hat{x}_B$ and $\hat{y}_A$ is $(90-\theta)$, and that between $\hat{y}_B$ and $\hat{y}_A$ is θ.

Rotation and translation in 2D

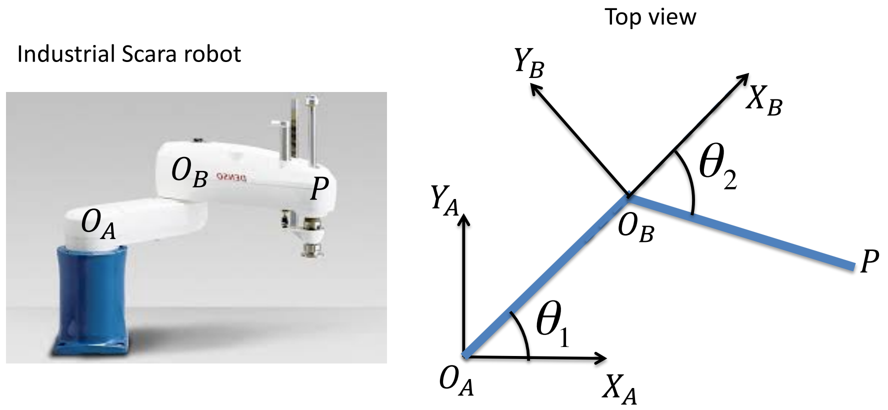


Figure 1.8: 2D rotation and translation in a Scara robot.

Figure 1.8 shows a scenario where we will have to use 2D translation and rotation to transform the vector $o_B P$ from frame B to frame A.

Figure 1.9 shows the rotation and translation in 2D in a general case where frame B is rotated by θ angle relative to frame A.

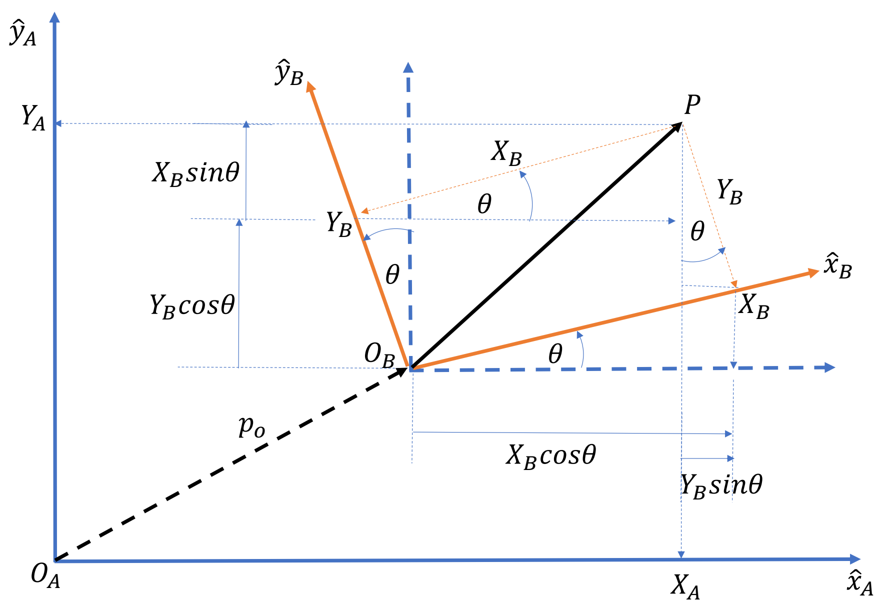


Figure 1.9: 2D rotation and translation.

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} & \begin{bmatrix} P_{xO} \\ P_{yO} \end{bmatrix} \\ 0 \quad\quad 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix}$$

Transformation matrix:

$$ {}^{A}\boldsymbol{T}_{B} = \begin{bmatrix} R_{11} & R_{12} & P_{xO} \\ R_{21} & R_{22} & P_{yO} \\ 0 & 0 & 1 \end{bmatrix} \tag{1.8} $$

1.4.2 Rotation and translation in 3D

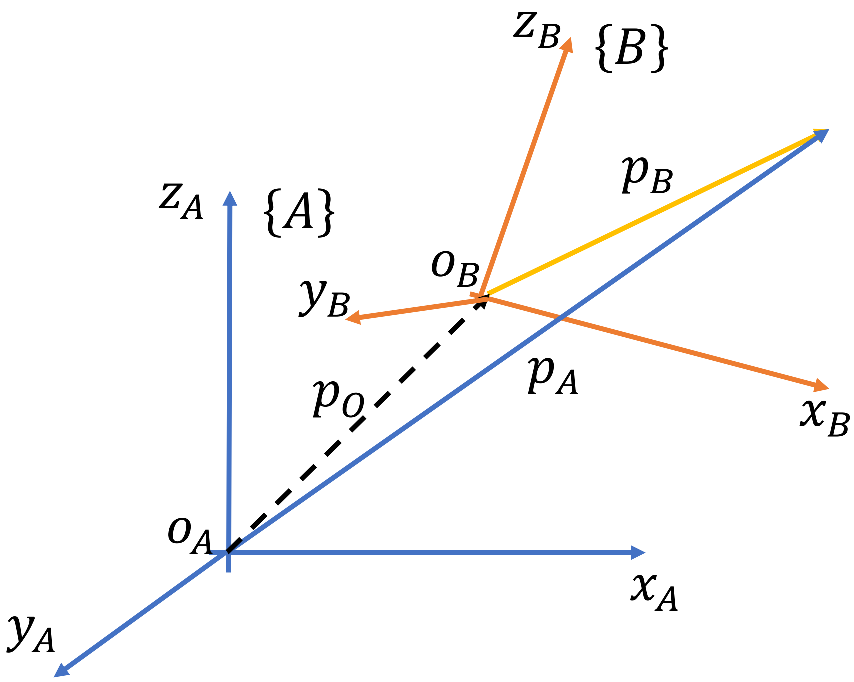


Figure 1.10: 3D translation and rotation of frames

Given vectors

$$ {}^{A}\boldsymbol{p} = \begin{bmatrix} p_{xA} \\ p_{yA} \\ p_{zA} \end{bmatrix}, \quad {}^{A}\boldsymbol{p}_{B} = \begin{bmatrix} p_{xO} \\ p_{yO} \\ p_{zO} \end{bmatrix}, \quad {}^{B}\boldsymbol{p} = \begin{bmatrix} p_{xB} \\ p_{yB} \\ p_{zB} \end{bmatrix} \tag{1.9} $$

The homogeneous coordinate transformation is given by

$$ \begin{bmatrix} p_{xA} \\ p_{yA} \\ p_{zA} \\ 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & p_{xO} \\ R_{21} & R_{22} & R_{23} & p_{yO} \\ R_{31} & R_{32} & R_{33} & p_{zO} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_{xB} \\ p_{yB} \\ p_{zB} \\ 1 \end{bmatrix} \tag{1.10} $$

$$ \begin{bmatrix} {}^{A}\boldsymbol{p} \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}\boldsymbol{R}_{B} & {}^{A}\boldsymbol{p}_{B} \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} {}^{B}\boldsymbol{p} \\ 1 \end{bmatrix} \tag{1.11} $$

where the transformation matrix:

$$ {}^{A}\boldsymbol{T}_{B} = \begin{bmatrix} {}^{A}\boldsymbol{R}_{B} & {}^{A}\boldsymbol{p}_{B} \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & p_{xO} \\ R_{21} & R_{22} & R_{23} & p_{yO} \\ R_{31} & R_{32} & R_{33} & p_{zO} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1.12} $$

Figure 1.10 shows how we can extend the idea of 2D rotation and translation to 3D. Now imagine what your brain might be doing when you recall walking inside a building, climbing up stairs, walking along corridors, taking left/right turns, etc., when the visual memories were collected in an eye-centered coordinate system. Think about how frame transformation should happen when you replay the memories to construct one integrated memory of walking inside a 3D space.

Rotation matrix from frame $(\hat{x}_j, \hat{y}_j, \hat{z}_j)$ to frame $(\hat{x}_i, \hat{y}_i, \hat{z}_i)$ is given by

$$ {}^{i}\boldsymbol{R}_j = \begin{bmatrix} \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{x}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{x}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{x}}_i \\ \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{y}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{y}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{y}}_i \\ \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{z}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{z}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{z}}_i \end{bmatrix} \tag{1.13}$$

where $(\hat{\boldsymbol{x}}_i, \hat{\boldsymbol{y}}_i, \hat{\boldsymbol{z}}_i)$ are the basis vectors (unit vectors along each axis are basis vectors) of frame i, and $(\hat{\boldsymbol{x}}_j, \hat{\boldsymbol{y}}_j, \hat{\boldsymbol{z}}_j)$ are those of frame j.

Rotation around $\hat{z}$ axis

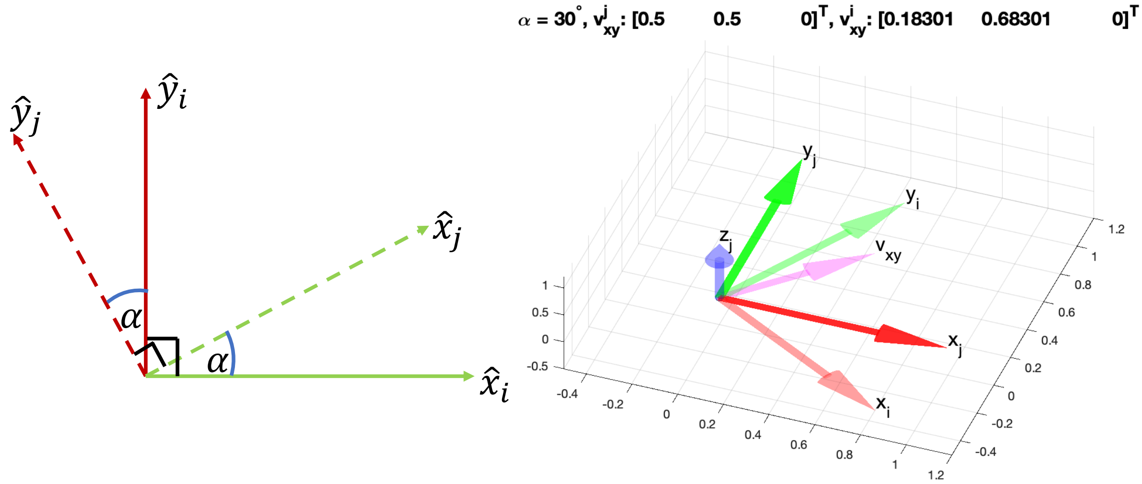


Figure 1.11: Frame rotation around $\hat{z}$ axis. Check z-axis section of ZYXindividualRotations.m for a 30° rotation.

$$ {}^{i}\boldsymbol{R}_j(\alpha) = \begin{bmatrix} |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{x}}_i|\cos(\alpha) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{x}}_i|\cos(90+\alpha) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{x}}_i|\cos(90) \\ |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{y}}_i|\cos(90-\alpha) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{y}}_i|\cos(\alpha) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{y}}_i|\cos(90) \\ |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{z}}_i|\cos(90) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{z}}_i|\cos(90) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{z}}_i|\cos(0) \end{bmatrix} \tag{1.14}$$

$$ \boldsymbol{R}_{\hat{z}}(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1.15}$$

Rotation around $\hat{y}$ axis.

$$ {}^{i}\boldsymbol{R}_j(\beta) = \begin{bmatrix} |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{x}}_i|\cos(\beta) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{x}}_i|\cos(90) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{x}}_i|\cos(90-\beta) \\ |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{y}}_i|\cos(90) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{y}}_i|\cos(0) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{y}}_i|\cos(90) \\ |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{z}}_i|\cos(90+\beta) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{z}}_i|\cos(90) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{z}}_i|\cos(\beta) \end{bmatrix} \tag{1.16}$$

$$ \boldsymbol{R}_{\hat{y}}(\beta) = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \tag{1.17}$$

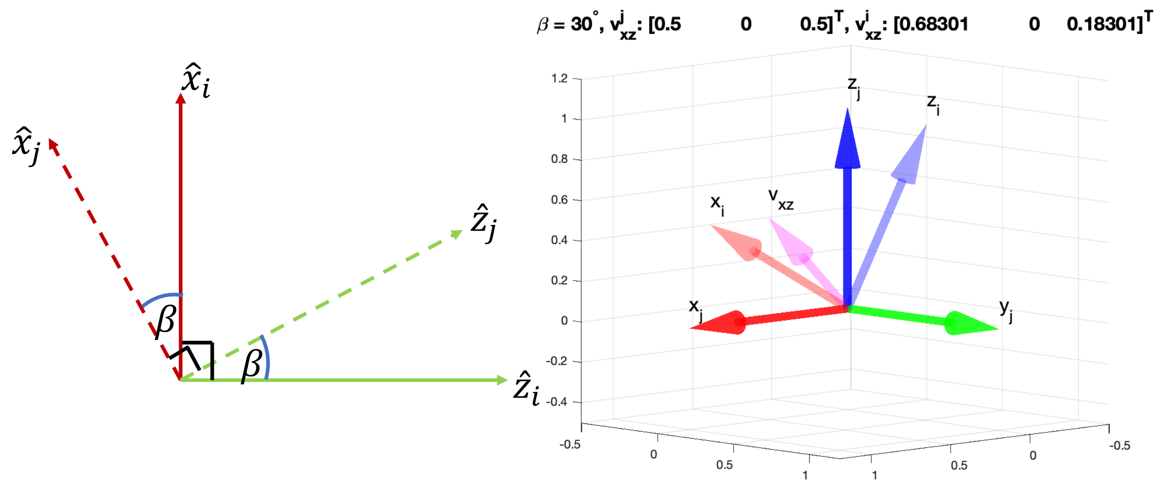


Figure 1.12: Frame rotation around $\hat{y}$ axis. Check y-axis section of ZYXindividualRotations.m for a 30° rotation.

Rotation around $\hat{x}$ axis

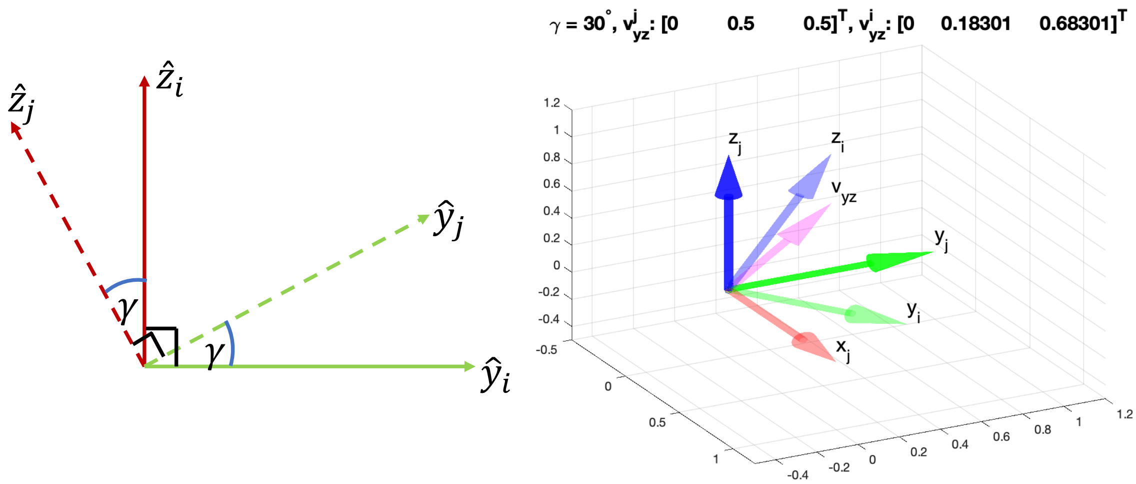


Figure 1.13: Frame rotation around $\hat{x}$ axis. Check x-axis section of ZYXindividualRotations.m for a 30° rotation.

$$
{}^{i}\boldsymbol{R}_j(\gamma) = \begin{bmatrix} |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{x}}_i|\cos(0) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{x}}_i|\cos(90) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{x}}_i|\cos(90) \\ |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{y}}_i|\cos(90) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{y}}_i|\cos(\gamma) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{y}}_i|\cos(90+\gamma) \\ |\hat{\boldsymbol{x}}_j||\hat{\boldsymbol{z}}_i|\cos(90) & |\hat{\boldsymbol{y}}_j||\hat{\boldsymbol{z}}_i|\cos(90-\gamma) & |\hat{\boldsymbol{z}}_j||\hat{\boldsymbol{z}}_i|\cos(\gamma) \end{bmatrix} \tag{1.18}
$$

$$
\boldsymbol{R}_{\hat{x}}(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix} \tag{1.19}
$$

1.4.3 Z-Y-X Euler angles: Rotation around $\hat{z}$, $\hat{y}$ and $\hat{x}$ axes

Leonard Euler said - "Any rotation can be described by three successive rotations around linearly independent axes". You can view each axis as a vector. When two vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$

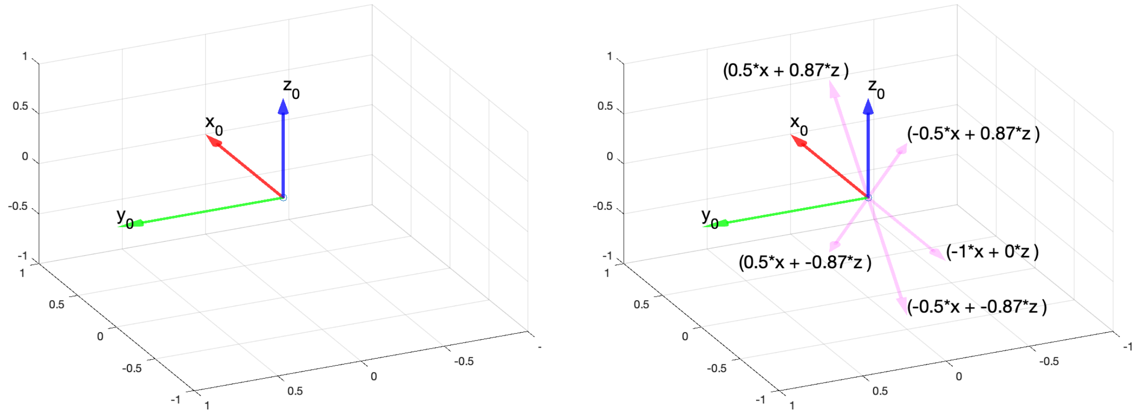


Figure 1.14: Idea of linear combinations of vectors

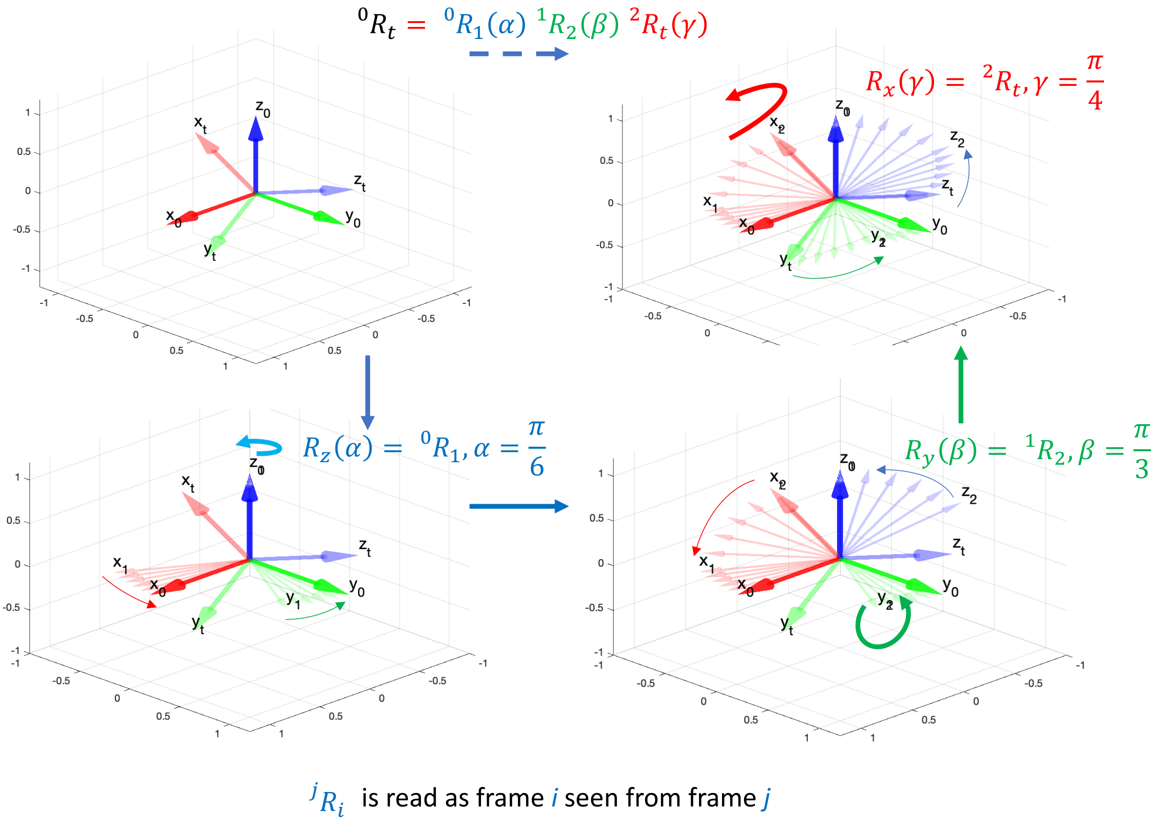


Figure 1.15: Z-Y-X Euler angles

are linearly combined in the form $\boldsymbol{v} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$, where c_1 and c_2 are scalar coefficients, $\boldsymbol{v}$ will be in the same plane of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ as shown in figure 1.14. Any vector outside this plane will be linearly independent of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$.

The objective of the first two Euler rotations with moving frames is to get one axis to coincide with the corresponding axis of the target frame. Then, since the three axes are orthogonal, one final rotation around the axis that coincides with the target axis will make all three axes to coincide.

A vector ${}^B\boldsymbol{p}$ in frame $\{B\}$ can be rotated to ${}^A\boldsymbol{p}$ in frame $\{A\}$ by a general Euler rotation given by

${}^{A}\boldsymbol{p} = \boldsymbol{q}\ {}^{B}\boldsymbol{p}\ \boldsymbol{q}^*$ where $\boldsymbol{q} = e^{\frac{\theta}{2}\boldsymbol{u}}$ and $\boldsymbol{q}^* = e^{-\frac{\theta}{2}\boldsymbol{u}}$, where $\boldsymbol{u}$ is a unit vector.

For instance, a rotation of a 2D vector in $X - Y$ plane around the z axis is given by $\boldsymbol{u} = 0\ i + 0\ j + k$.

$$\begin{aligned} {}^{A}\boldsymbol{p} &= \boldsymbol{q}\ {}^{B}\boldsymbol{p}\ \boldsymbol{q}^* \\ &= \left(\cos\frac{\theta}{2} + k\sin\frac{\theta}{2}\right)\left({}^{B}\boldsymbol{p}_x\ i + {}^{B}\boldsymbol{p}_y\ j + 0\ k\right)\left(\cos\frac{\theta}{2} - k\sin\frac{\theta}{2}\right) \end{aligned} \tag{1.20}$$

$$\begin{aligned} {}^{A}v &= \boldsymbol{q}\ {}^{B}v\ \boldsymbol{q}^* \\ &= \left(\cos\frac{\theta}{2} + k\sin\frac{\theta}{2}\right)\left({}^{B}v_x\ i + {}^{B}v_y\ j\right)\left(\cos\frac{\theta}{2} - k\sin\frac{\theta}{2}\right) \end{aligned} \tag{1.21}$$

By expanding, we obtain

$$\begin{aligned} {}^{A}v &= \left(\cos\frac{\theta}{2} + k\sin\frac{\theta}{2}\right)\left({}^{B}v_x\ i + {}^{B}v_y\ j\right)\left(\cos\frac{\theta}{2} - k\sin\frac{\theta}{2}\right) \\ &= i\ \left({}^{B}v_x\left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right) - 2\ {}^{B}v_y \sin\frac{\theta}{2}\cos\frac{\theta}{2}\right) \\ &+ j\ \left({}^{B}v_y\left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right) + 2\ {}^{B}v_x \sin\frac{\theta}{2}\cos\frac{\theta}{2}\right) \\ &= i\ \left({}^{B}v_x\cos\theta - {}^{B}v_y\sin\theta\right) + j\ \left({}^{B}v_x\sin\theta + {}^{B}v_y\cos\theta\right) \end{aligned} \tag{1.22}$$

We can then re-arrange the terms to form the rotation matrix given by

$$\begin{bmatrix} {}^{A}v_x \\ {}^{A}v_y \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} {}^{B}v_x \\ {}^{B}v_y \end{bmatrix} \tag{1.23}$$

In homogeneous coordinates,

$$\begin{bmatrix} {}^{A}v_x \\ {}^{A}v_y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}v_x \\ {}^{B}v_y \\ 1 \end{bmatrix} \tag{1.24}$$

In $Z - Y - X$ Euler rotations, frame j is rotated by α around $\hat{\boldsymbol{z}}_0$ to get $\hat{\boldsymbol{x}}_1$ and $\hat{\boldsymbol{x}}_t$ to be on the plane perpendicular to $\hat{\boldsymbol{y}}_1$. Then, a rotation around the new $\hat{\boldsymbol{y}}_1$ by an angle β will get the two x-axes to coincide. Since this automatically makes the $Y - Z$ planes coincide, a final γ rotation around the new $\hat{\boldsymbol{x}}_2$ axis to finally get all three axes coincide with frame i.

Once we have tracked how the rotations should be done, we implement them in reverse order where we first perform the rotation ${}^{2}\boldsymbol{R}_t$ (read as a rotation from frame t to frame 2), then ${}^{1}\boldsymbol{R}_2$, and finally ${}^{0}\boldsymbol{R}_1$. Then the compound transformation matrix ${}^{0}\boldsymbol{R}_t$ is given by

$${}^{0}\boldsymbol{R}_t = {}^{0}\boldsymbol{R}_1\ {}^{1}\boldsymbol{R}_2\ {}^{2}\boldsymbol{R}_t \tag{1.25}$$

$${}^{0}\boldsymbol{R}_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

Run visualizeZYXEuler.m

$$
\begin{aligned}
{}^1\boldsymbol{R}_t(\alpha,\beta,\gamma) &= \begin{bmatrix} c\,(\alpha)\; c\,(\beta) & -c\,(\gamma)\; s\,(\alpha) + c\,(\alpha)\; s\,(\beta)\; s\,(\gamma) & s\,(\alpha)\; s\,(\gamma) + c\,(\alpha)\; c\,(\gamma)\; s\,(\beta) \\ c\,(\beta)\; s\,(\alpha) & c\,(\alpha)\; c\,(\gamma) + s\,(\alpha)\; s\,(\beta)\; s\,(\gamma) & -c\,(\alpha)\; s\,(\gamma) + c\,(\gamma)\; s\,(\alpha)\; s\,(\beta) \\ -s\,(\beta) & c\,(\beta)\; s\,(\gamma) & c\,(\beta)\; c\,(\gamma) \end{bmatrix} \\
&= \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \qquad (1.26)
\end{aligned}
$$

where, the short forms c and s are for cos and sin respectively.

Given a rotation matrix (run zyxEuler.m), can we compute the Euler angles?

We can use $R_{31} = -s(\beta)$ to compute $\beta = -\arcsin(R_{31})$. Then we can use $R_{11} = c(\alpha)c(\beta)$ and $R_{21} = c(\beta)s(\alpha)$ to compute α using $\alpha = atan2(\frac{R_{21}}{c\beta}, \frac{R_{11}}{c\beta})$. Similarly we can compute $\gamma = atan2(\frac{R_{32}}{c\beta}, \frac{R_{33}}{c\beta})$.

Can you compute the angle between the z_1 and z_t axes?

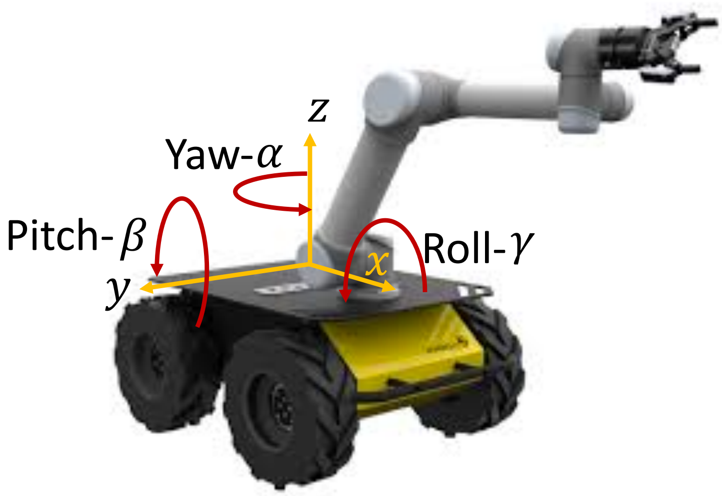
Answer: Let angle between the z_1 and z_t axes be ψ. Then $\psi = \arccos(c\beta c\gamma)$.

Can you guess how many patterns we can use to perform Euler rotations?

X-Y-Z Fixed Angle Rotations (Roll, Yaw, and Pitch)

There are 3 successive rotations around the original axes as opposed to new axes. Roll, Yaw, and Pitch are often used to denote orientation of robotic links.

- Roll: Rotation around the x-axis (γ)
- Yaw: Rotation around the z-axis (α)
- Pitch: Rotation around the y-axis (β)



$$
{}^0\boldsymbol{R}_t(\gamma,\beta,\alpha) = \; \boldsymbol{R}_{z_0}(\alpha)\; \boldsymbol{R}_{y_0}(\beta)\; \boldsymbol{R}_{x_0}(\gamma) \qquad (1.27)
$$

$$
{}^1\boldsymbol{R}_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}
$$

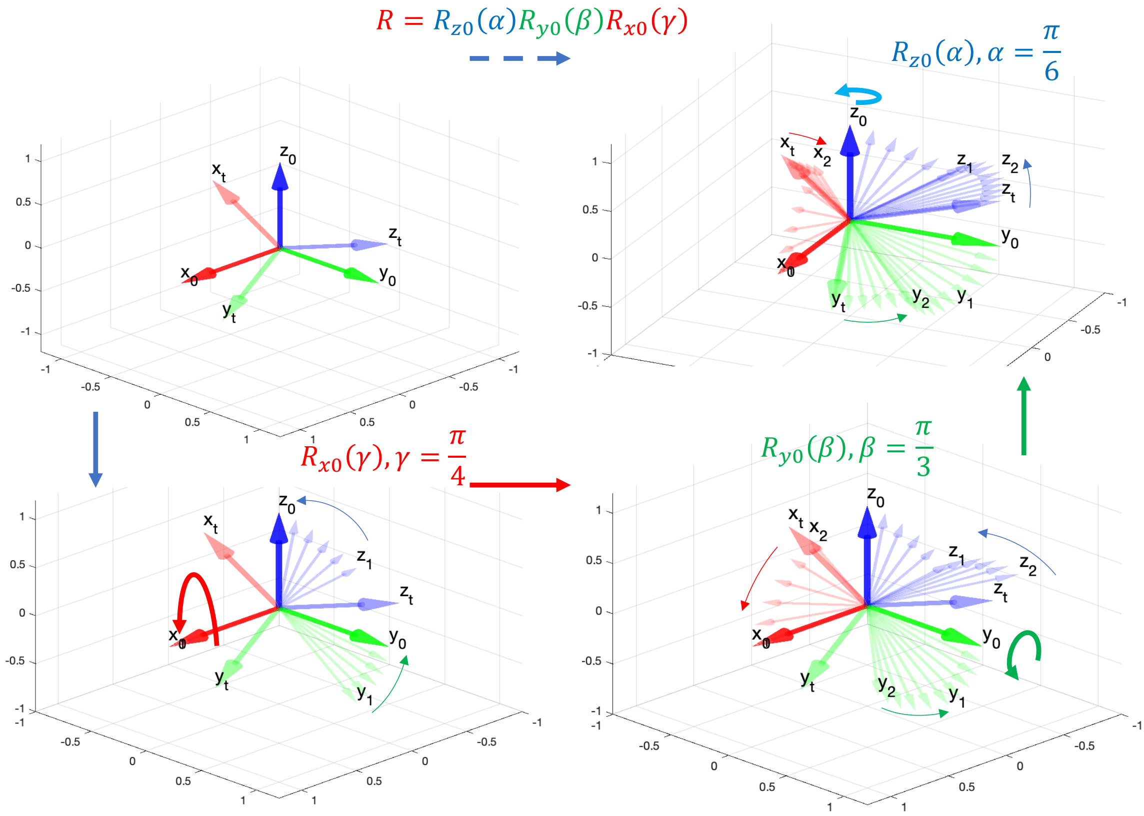


Figure 1.16: X-Y-Z Fixed rotations

Run visualizeZYXfixed.m

Which gives

$$
\begin{aligned}
{}^1\boldsymbol{R}_t(\alpha,\beta,\gamma) &= \begin{bmatrix} c(\alpha)\ c(\beta) & -c(\gamma)\ s(\alpha) + c(\alpha)\ s(\beta)\ s(\gamma) & s(\alpha)\ s(\gamma) + c(\alpha)\ c(\gamma)\ s(\beta) \\ c(\beta)\ s(\alpha) & c(\alpha)\ c(\gamma) + s(\alpha)\ s(\beta)\ s(\gamma) & -c(\alpha)\ s(\gamma) + c(\gamma)\ s(\alpha)\ s(\beta) \\ -s(\beta) & c(\beta)\ s(\gamma) & c(\beta)\ c(\gamma) \end{bmatrix} \\
&= \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \qquad (1.28)
\end{aligned}
$$

Rotation with quaternions

The idea behind quaternions is that we can find a unit vector

$$\boldsymbol{u} = k_1 i + k_2 j + k_3 k$$

called the Euler axis in the base frame such that a rotation by an angle θ around $\boldsymbol{u}$ achieves a 3D rotation. Run Quaternion.m

This can be written in the form of an Euler rotation of a vector p in the base frame to $p^{'}$ given by

$$p^{'} = \boldsymbol{q}\ p\ \boldsymbol{q}^*$$

where

$$\boldsymbol{q} = e^{\frac{\theta}{2}\boldsymbol{u}}$$

and

$$q^* = e^{-\frac{\theta}{2}\boldsymbol{u}}$$

.

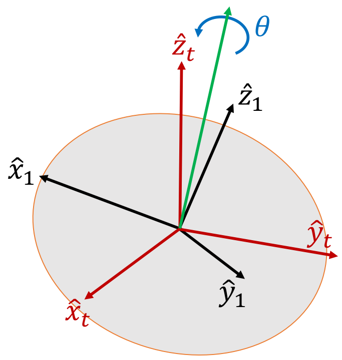
$\boldsymbol{q}^*$ is called the conjugate of $\boldsymbol{q}$.

$$\begin{aligned} \boldsymbol{q} &= e^{\frac{\theta}{2}\boldsymbol{u}} \\ &= e^{\frac{\theta}{2}(k_1 i + k_2 j + k_3 k)} \\ &= \cos\frac{\theta}{2} + (k_1 i + k_2 j + k_3 k)\sin\frac{\theta}{2} \end{aligned} \tag{1.29}$$

Vector $\boldsymbol{q}$ can be written as

$$\boldsymbol{q} = \epsilon_1 i + \epsilon_2 j + \epsilon_3 k + \epsilon_4$$

in the base frame so that a rotation of the base frame around vector $\boldsymbol{u}$ by an angle θ will bring it to coincide with the target frame.



We define four parameters called Euler parameters: $\epsilon_1 = k_1 \sin\frac{\theta}{2}$, $\epsilon_2 = k_2 \sin\frac{\theta}{2}$, $\epsilon_3 = k_3 \sin\frac{\theta}{2}$, and $\epsilon_4 = \cos\frac{\theta}{2}$, where $\hat{K} = [k_1,\ k_2,\ k_3]^T$ is a unit vector.

This gives a constraint

$$\begin{aligned} \epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 &= \left(k_1^2 + k_2^2 + k_3^2\right)\sin^2\frac{\theta}{2} + \cos^2\frac{\theta}{2} \\ &= 1 \end{aligned} \tag{1.30}$$

because $k_1^2 + k_2^2 + k_3^2 = 1$ for the unit vector $\hat{K}$. Since $\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 = 1$, the 4 element vector $[\epsilon_1,\ \epsilon_2,\ \epsilon_3,\ \epsilon_4]$ is called the unit quaternion.

Then a 3D rotation around $\boldsymbol{u}$ vector by an angle θ is given by

$$\begin{aligned} {}^A v &= R_\epsilon\ {}^B v \\ &= \boldsymbol{q}\ {}^B v\ \boldsymbol{q}^* \end{aligned} \tag{1.31}$$

where,

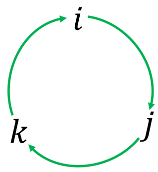
$$\boldsymbol{q} = \epsilon_1 i + \epsilon_2 j + \epsilon_3 k + \epsilon_4$$

$$\boldsymbol{q}^* = -\epsilon_1 i - \epsilon_2 j - \epsilon_3 k + \epsilon_4$$

and

$${}^B v = {}^B v_x\ i + {}^B v_y\ j + {}^B v_z\ k$$

Please note $i*i=j*j=k*k=-1$, $i*j=k$, $j*k=i$, $k*i=j$, and $j*i=-k$, $k*j=-i$, $i*k=-j$.



Then the rotation matrix is given by

$$R_\epsilon = \begin{bmatrix} 1-2\epsilon_2^2-2\epsilon_3^2 & 2\left(\epsilon_1\epsilon_2-\epsilon_3\epsilon_4\right) & 2\left(\epsilon_1\epsilon_3+\epsilon_2\epsilon_4\right) \\ 2\left(\epsilon_1\epsilon_2+\epsilon_3\epsilon_4\right) & 1-2\epsilon_1^2-2\epsilon_3^2 & 2\left(\epsilon_2\epsilon_3-\epsilon_1\epsilon_4\right) \\ 2\left(\epsilon_1\epsilon_3-\epsilon_2\epsilon_4\right) & 2\left(\epsilon_2\epsilon_3+\epsilon_1\epsilon_4\right) & 1-2\epsilon_1^2-2\epsilon_2^2 \end{bmatrix}$$

We notice that

$$\begin{aligned} R_{11}+R_{22}+R_{33} &= \left(1-2\epsilon_2^2-2\epsilon_3^2\right)+\left(1-2\epsilon_1^2-2\epsilon_3^2\right)+\left(1-2\epsilon_1^2-2\epsilon_2^2\right) \\ &= 3-4\epsilon_1^2-4\epsilon_2^2-4\epsilon_3^2 \end{aligned} \tag{1.32}$$

By adding a 1 on both sides, we obtain

$$\begin{aligned} 1+R_{11}+R_{22}+R_{33} &= 4-4\epsilon_1^2-4\epsilon_2^2-4\epsilon_3^2 \\ &= 4\left(1-\left(\epsilon_1^2+\epsilon_2^2+\epsilon_3^2\right)\right) \end{aligned} \tag{1.33}$$

Substituting ϵ_4^2 from equation 1.30, we obtain

$$\begin{aligned} 1+R_{11}+R_{22}+R_{33} &= 4\left(1-\left(1-\epsilon_4^2\right)\right) \\ &= 4\epsilon_4^2 \end{aligned} \tag{1.34}$$

Therefore, $\epsilon_4 = \frac{1}{2}\sqrt{1+R_{11}+R_{22}+R_{33}}$.
Therefore, the Euler parameters are given by

$$\epsilon_1 = \frac{R_{32}-R_{23}}{4\epsilon_4} \tag{1.35}$$

$$\epsilon_2 = \frac{R_{13}-R_{31}}{4\epsilon_4} \tag{1.36}$$

$$\epsilon_3 = \frac{R_{21}-R_{12}}{4\epsilon_4} \tag{1.37}$$

and

$$\epsilon_4 = \frac{1}{2}\sqrt{1+R_{11}+R_{22}+R_{33}} \tag{1.38}$$

Let us take a 2D rotation example.
In this case, the unit quaternion vector becomes

$$\begin{aligned} \boldsymbol{q} &= \epsilon_4 + k\epsilon_3 \\ &= \cos\frac{\theta}{2} + k\sin\frac{\theta}{2} \end{aligned} \tag{1.39}$$

because $k_1=0$, $k_2=0$, and $k_3=1$ to satisfy $k_1^2+k_2^2+k_3^2=1$.
Then, given

$${}^B v = {}^B v_x\ i + {}^B v_y\ j$$

Then

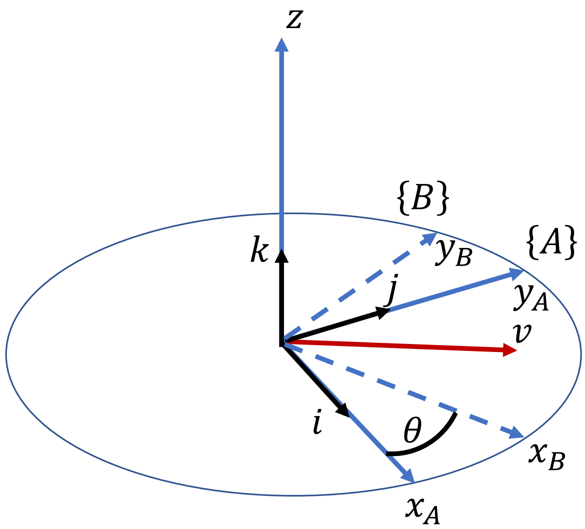


Figure 1.17: Example of a 2D rotation around the z axis.

$$
\begin{aligned}
{}^{A}v &= \boldsymbol{q}\ {}^{B}v\ \boldsymbol{q}^{*} \\
&= \left(\cos\frac{\theta}{2} + k\sin\frac{\theta}{2}\right)\left({}^{B}v_x\ i +{}^{B} v_y\ j\right)\left(\cos\frac{\theta}{2} - k\sin\frac{\theta}{2}\right)
\end{aligned} \tag{1.40}
$$

By expanding, we obtain

$$
\begin{aligned}
{}^{A}v &= \left(\cos\frac{\theta}{2} + k\sin\frac{\theta}{2}\right)\left({}^{B}v_x\ i +\ {}^{B}v_y\ j\right)\left(\cos\frac{\theta}{2} - k\sin\frac{\theta}{2}\right) \\
&= \left(i\ {}^{B}v_x\cos\frac{\theta}{2} + j\ {}^{B}v_y\cos\frac{\theta}{2} + j\ {}^{B}v_x\sin\frac{\theta}{2} - i\ {}^{B}v_y\sin\frac{\theta}{2}\right)\left(\cos\frac{\theta}{2} - k\sin\frac{\theta}{2}\right) \\
&= i\ {}^{B}v_x\cos^2\frac{\theta}{2} + j\ {}^{B}v_x\cos\frac{\theta}{2}\sin\frac{\theta}{2} \\
&+ j\ {}^{B}v_y\cos^2\frac{\theta}{2} - i\ {}^{B}v_y\cos\frac{\theta}{2}\sin\frac{\theta}{2} \\
&+ j\ {}^{B}v_x\sin\frac{\theta}{2}\cos\frac{\theta}{2} - i\ {}^{B}v_x\sin^2\frac{\theta}{2} \\
&+ -i\ {}^{B}v_y\sin\frac{\theta}{2}\cos\frac{\theta}{2} - j\ {}^{B}v_y\sin^2\frac{\theta}{2} \\
&= i\ \left({}^{B}v_x\left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right) - 2\ {}^{B}v_y\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right) \\
&+ j\ \left({}^{B}v_y\left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right) + 2\ {}^{B}v_x\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right) \\
&= i\ \left({}^{B}v_x\cos\theta -{}^{B} v_y\sin\theta\right) + j\ \left({}^{B}v_x\sin\theta +{}^{B} v_y\cos\theta\right)
\end{aligned} \tag{1.41}
$$

Note: $\cos(\frac{\theta}{2}+\frac{\theta}{2}) = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$ and $\sin(\frac{\theta}{2}+\frac{\theta}{2}) = \sin\frac{\theta}{2}\cos\frac{\theta}{2} + \cos\frac{\theta}{2}\sin\frac{\theta}{2} = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$.

Then, we can then re-arrange the terms to form the rotation matrix given by

$$
\begin{bmatrix} {}^{A}v_x \\ {}^{A}v_y \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}\begin{bmatrix} {}^{B}v_x \\ {}^{B}v_y \end{bmatrix} \tag{1.42}
$$

Properties of transformation matrices

Property-1: The inverse of a transformation matrix is also a transformation matrix and has the following form.

$$\boldsymbol{T}^{-1} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{P} \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{R}^T & -\boldsymbol{R}^T\,\boldsymbol{P} \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix} \tag{1.43}$$

Note that this property allows us to avoid doing computationally heavy matrix inversions. This is very useful for realtime applications where computational speed matters.

Property-2: The product of two transformation matrices is also a transformation matrix.

$$\boldsymbol{T}_1\boldsymbol{T}_2 = \begin{bmatrix} \boldsymbol{R}_1 & \boldsymbol{P}_1 \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix}\begin{bmatrix} \boldsymbol{R}_2 & \boldsymbol{P}_2 \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}_1\boldsymbol{R}_2 & \boldsymbol{R}_1\boldsymbol{P}_2 + \boldsymbol{P}_1 \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix} \tag{1.44}$$

This property allows us to break down a frame transformation into intermediate steps or via points to simplify the calculations. We can obtain the compound transformation matrix by multiplying the intermediate transformation matrices in a series.

Property-3: Multiplication of transformation matrices is associative, $(\boldsymbol{T}_1\,\boldsymbol{T}_2)\,\boldsymbol{T}_3 = \boldsymbol{T}_1\,(\boldsymbol{T}_2\,\boldsymbol{T}_3)$, but generally not commutative, $\boldsymbol{T}_1\,\boldsymbol{T}_2 \neq \boldsymbol{T}_2\,\boldsymbol{T}_1$

Compound transformations

Example: ${}^B\boldsymbol{P}_C$ is read as the vector to the origin of frame C in frame B.

$$\begin{aligned} {}^A\boldsymbol{T}_D &= {}^A\boldsymbol{T}_B\;{}^B\boldsymbol{T}_C\;{}^C\boldsymbol{T}_D \qquad (1.45)\\ &= \begin{bmatrix} {}^A\boldsymbol{R}_B & {}^A\boldsymbol{P}_B \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix}\begin{bmatrix} {}^B\boldsymbol{R}_C & {}^B\boldsymbol{P}_C \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix}\begin{bmatrix} {}^C\boldsymbol{R}_D & {}^C\boldsymbol{P}_D \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix} \end{aligned}$$

$${}^A\boldsymbol{T}_D = \begin{bmatrix} {}^A\boldsymbol{R}_B\;{}^B\boldsymbol{R}_C\;{}^C\boldsymbol{R}_D & \left({}^A\boldsymbol{R}_B\;{}^B\boldsymbol{R}_C\;{}^C\boldsymbol{P}_D +^A \boldsymbol{R}_B\;{}^B\boldsymbol{P}_C +^A \boldsymbol{P}_B\right) \\ \boldsymbol{O}_{1\times3} & 1 \end{bmatrix} \tag{1.46}$$

1.5 Jacobian matrix from the position vector

Once we know the homogeneous transformation matrix for a robot manipulator, we can obtain the rotation matrix which gives the orientation of the last link relative to the base link, and the position of the end effector relative to the base coordinate system.

A matrix known as the Jacobian matrix can be derived from the position vector to solve the inverse kinematics problem.
Let the position vector of the end effector of a robot manipulator be given by $\boldsymbol{p} = f(\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is the joint angle vector. Then we can use the chain rule to obtain the speed vector of the end effector with respective to the joint speeds given by

$$\frac{d\boldsymbol{p}}{dt} = \frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\frac{d\boldsymbol{\theta}}{dt} \tag{1.47}$$

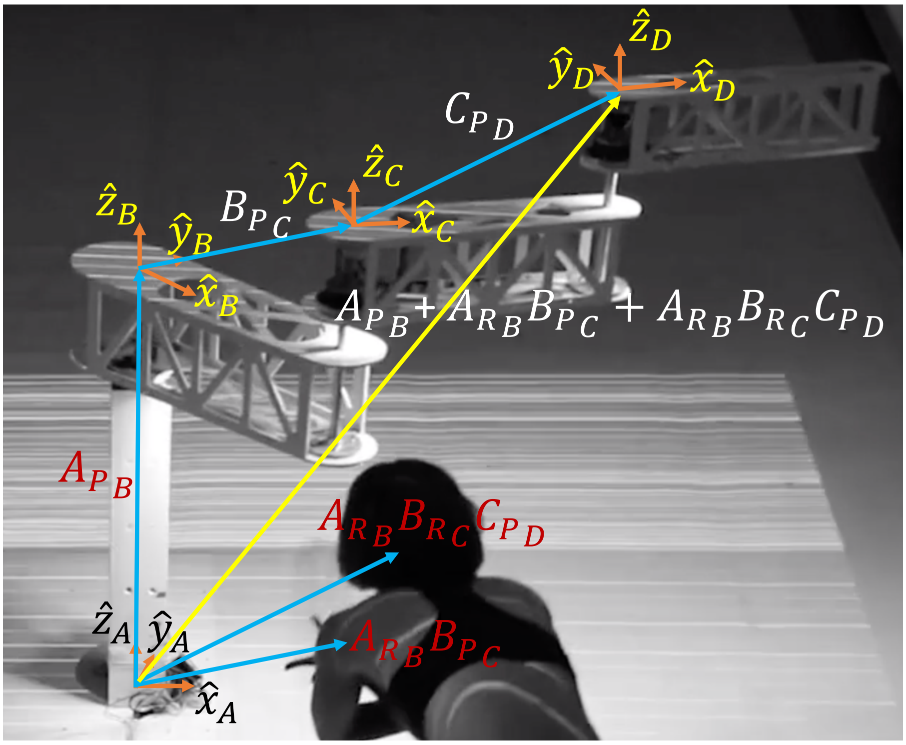


Figure 1.18: Compound Transformation

Then the Jacobian matrix

$$J = \frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \tag{1.48}$$

Please note that $f(\boldsymbol{\theta})$ is a vector of functions of $\boldsymbol{\theta}$, such as

$$f(\boldsymbol{\theta}) = \begin{bmatrix} p_x(\boldsymbol{\theta}) \\ p_y(\boldsymbol{\theta}) \\ p_z(\boldsymbol{\theta}) \\ \alpha(\boldsymbol{\theta}) \\ \beta(\boldsymbol{\theta}) \\ \gamma(\boldsymbol{\theta}) \end{bmatrix} \tag{1.49}$$

where $p_x(\boldsymbol{\theta})$, $p_y(\boldsymbol{\theta})$, and $p_y(\boldsymbol{\theta})$ are x, y, and z components of the position vector, and $\alpha(\boldsymbol{\theta})$, $\beta(\boldsymbol{\theta})$, and $\gamma(\boldsymbol{\theta})$ are yaw, pitch, and roll respectively. Please also note that there can be user defined kinematic functions in the $f(\boldsymbol{\theta})$ vector. In some industry scenarios, a user can define constraints as user defined kinematic functions. This is common in robot manipulators with redundant degrees of freedom. It also helps to make the Jacobian matrix a square matrix.

Physical meaning of the columns of the Jacobian matrix

Figure 1.19 shows the physical meaning of the column vectors of the Jacobian matrix. In this case, for simplicity, we consider only the position vector in $f(\boldsymbol{\theta})$ given by

$$f(\boldsymbol{\theta}) = \begin{bmatrix} p_x(\boldsymbol{\theta}) \\ p_y(\boldsymbol{\theta}) \\ p_z(\boldsymbol{\theta}) \end{bmatrix} \tag{1.50}$$

Then the Jacobian is given by

$$J(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial p_x}{\partial \theta_1} & \frac{\partial p_x}{\partial \theta_2} & \cdots & \frac{\partial p_x}{\partial \theta_N} \\ \frac{\partial p_y}{\partial \theta_1} & \frac{\partial p_y}{\partial \theta_2} & \cdots & \frac{\partial p_y}{\partial \theta_N} \\ \frac{\partial p_z}{\partial \theta_1} & \frac{\partial p_z}{\partial \theta_2} & \cdots & \frac{\partial p_z}{\partial \theta_N} \end{bmatrix} \tag{1.51}$$

This means, each column of the Jacobian matrix is the gradient of the task space vector $f(\boldsymbol{\theta}) = \begin{bmatrix} p_x(\boldsymbol{\theta}) \\ p_y(\boldsymbol{\theta}) \\ p_z(\boldsymbol{\theta}) \end{bmatrix}$ along each joint angle. In other words, each column of the Jacobian matrix gives the direction the end point of the robot manipulator would move when each joint is rotated.

Then the end-effector velocity given by equation 1.52

$$\begin{aligned} \dot{\boldsymbol{p}} &= J\dot{\boldsymbol{\theta}} \\ v &= J\dot{\boldsymbol{\theta}} \end{aligned} \tag{1.52}$$

is a weighted sum of the columns of the Jacobian matrix scaled by the corresponding joint angular velocity. When a manipulator has more degrees of freedom, it has more vectors contributed by the Jacobian matrix to move the end-effector in the 3D space. We call them degrees of freedom.

When two columns of the Jacobian matrix become parallel to each other, the manipulator loses one degree of freedom to move the end-effector by actuating its joints. Effectively, two different columns become "singular". In other words, actuation of these two joints move the end- effector in the same direction. The end-effector still moves in 3D space as long as the manipulator has enough degrees of freedom left to move in the 3D space. However, if the number of non co-linear columns of the Jacobian matrix is less than 3, then the end-effector can not be moved in any desired direction in the 3D space. It will be limited to a plane or a line defined by the remaining non co-linear columns of the Jacobian matrix.

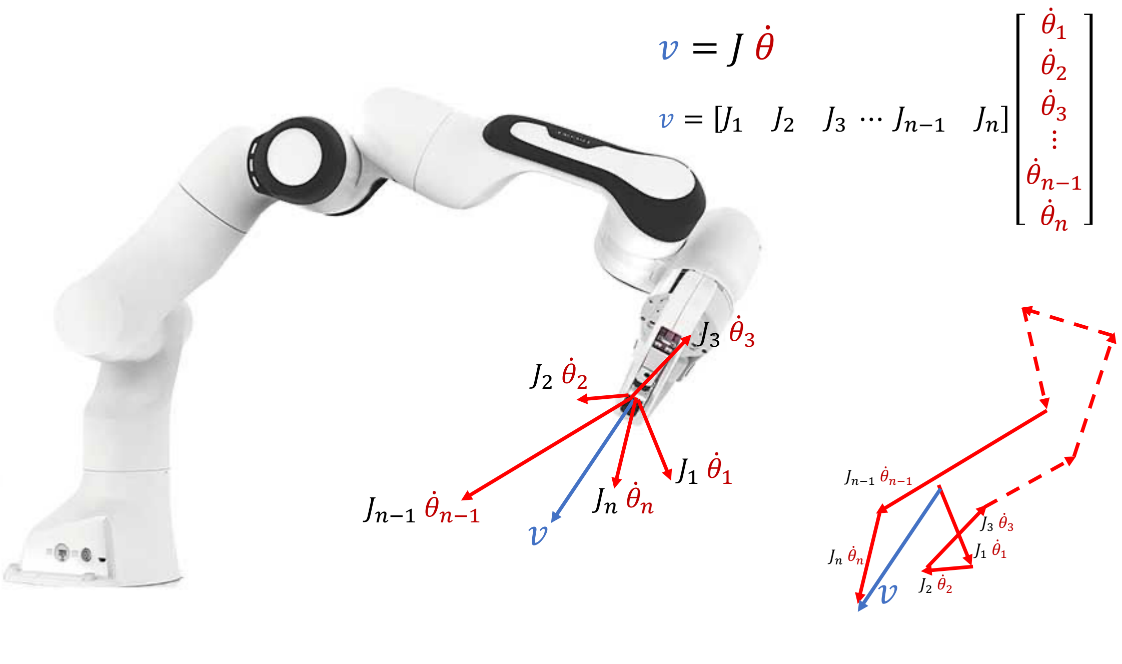


Figure 1.19: The physical meaning of the column vectors of the Jacobian matrix.

Please note that the idea of making a 3D movement of a point on the manipulator using a weighted sum of the columns of the Jacobian matrix applies not only to the end-effector, but also to the movement of any point on the manipulator. Therefore, the location of joints in the manipulator can be controlled by the Jacobian Matrix applicable to joints below it. Therefore, in a manipulator with redundant degrees of freedom (a manipulator having more degrees of freedom than what is required by the task at the end effector), the position of the 3rd joint and above can be moved in 3D space with at least 3 degrees of freedom unless the corresponding columns in the Jacobian matrix are not parallel. This allows a manipulator with redundant degrees of freedom to move the end effector in 3D space while controlling the shape of the manipulator too.

Computing the joint angular velocities required to move the end effector in 3D space

If J is a square matrix, we can obtain the joint angular speeds given an end-point speed vector using $\dot{\boldsymbol{\theta}} = J^{-1}\dot{\boldsymbol{p}}$. If this can be computed, a given desired end-effector position profile can first be differentiated to obtain the speed profile, and then the above relationship can be used to obtain the joint angular speed profile. Then the joint angle profile can be obtained by integrating the joint angular speed profile.
However, in most cases, the Jacobian matrix is not a square matrix due to more than 3-joints in the manipulator. In such cases, we use the Moore-Penrose pseudo-inverse of the Jacobian given by $J^+ = J^T(JJ^T)^{-1}$. If J is an $m \times n$ matrix such that $m < n$, then (JJ^T) is an $m \times m$ matrix, and hence $(JJ^T)^{-1}$ is also an $m \times m$ matrix. Since J^T is an $n \times m$ matrix, $J^T(JJ^T)^{-1}$ is an $n \times m$ matrix. Therefore, the pseudoinverse of a Jacobian matrix is of the dimensions of its transform.

1.5.1 Mapping external forces to joint torques

In the case of a robotic manipulator, the work done by the torques around joints for a small joint movement is equal to the work done by the force vectors at contact points on the displacements of those points.

Here, we use the concept of **virtual displacements**. A virtual displacement is a hypothetical (hence the term virtual) small movement from the actual trajectory or position/posture of a system without violating the system's constraints.

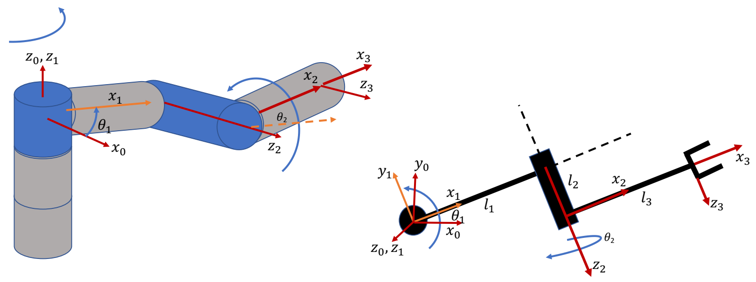
Virtual work is the work done by a force/torque vector on a virtual displacement.

Consider a vector of virtual joint movements $\boldsymbol{\delta\theta}$ that result in a virtual end point movement vector $\boldsymbol{\delta p}$. Then the virtual work done by the torque vector $\boldsymbol{\tau}$ on $\boldsymbol{\delta\theta}$ = virtual work done by a force vector $\boldsymbol{f}$ on a displacement vector $\delta\boldsymbol{p}$.

$$\begin{aligned}
\boldsymbol{\tau}^T\delta\boldsymbol{\theta} &= \boldsymbol{f}^T\delta\boldsymbol{p} \qquad (1.53)\\
\lim_{\delta t\to 0}\boldsymbol{\tau}^T\frac{\delta\boldsymbol{\theta}}{\delta t} &= \lim_{\delta t\to 0}\boldsymbol{f}^T\frac{\delta\boldsymbol{p}}{\delta t}\\
\boldsymbol{\tau}^T\dot{\boldsymbol{\theta}} &= \boldsymbol{f}^T\dot{\boldsymbol{p}}\\
\boldsymbol{\tau}^T\dot{\boldsymbol{\theta}} &= \boldsymbol{f}^T J\dot{\boldsymbol{\theta}}\\
\boldsymbol{\tau}^T &= \boldsymbol{f}^T J\\
\boldsymbol{\tau} &= J^T\boldsymbol{f}
\end{aligned}$$

(1.54)

1.5.2 Example 1



The homogeneous transformation matrix of a 3-link (2 joints) robot manipulator is given by ${}^0\boldsymbol{T}_3 = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2\,{}^2\boldsymbol{T}_3$

$$
{}^0\boldsymbol{T}_3 = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 & l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 & l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ s\theta_2 & c\theta_2 & 0 & l_3 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1.55}
$$

For unit link lengths, use the Pseudoinverse of the Jacobian matrix to obtain the joint angles and angular velocities for unit velocity vectors at the end point, for two initial configurations: $\theta_1 = \pi/6$, $\theta_2 = \pi/6$ and $\theta_1 = 0$, $\theta_2 = 0$.

Hence make your own observations about how the joint angle configuration affects the joint velocity vector for a given end-effector velocity vector.

We obtained that the position vector is given by

$$
{}^0\boldsymbol{P}_3 = \begin{bmatrix} l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ l_3 s\theta_2 \end{bmatrix} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \tag{1.56}
$$

Then the Jacobian is given by

$$
J(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial p_x}{\partial \theta_1} & \frac{\partial p_x}{\partial \theta_2} \\ \frac{\partial p_y}{\partial \theta_1} & \frac{\partial p_y}{\partial \theta_2} \\ \frac{\partial p_z}{\partial \theta_1} & \frac{\partial p_z}{\partial \theta_2} \end{bmatrix} \tag{1.57}
$$

Run JacobianMat.m to obtain the Jacobian matrix.

$$
J(\boldsymbol{\theta}) = \begin{bmatrix} l_2\cos(\theta_1) - l_1\sin(\theta_1) - l_3\cos(\theta_2)\sin(\theta_1), & -l_3\cos(\theta_1)\sin(\theta_2) \\ l_1\cos(\theta_1) + l_2\sin(\theta_1) + l_3\cos(\theta_1)\cos(\theta_2), & -l_3\sin(\theta_1)\sin(\theta_2) \\ 0, & l_3\cos(\theta_2) \end{bmatrix} \tag{1.58}
$$

$$
\dot{\boldsymbol{\theta}} = J^+(\boldsymbol{\theta})\dot{\boldsymbol{p}} \tag{1.59}
$$

1. $J(\boldsymbol{\theta})$ has elements that are functions of $\boldsymbol{\theta}$.

2. Then, the map from $\dot{\boldsymbol{p}}$ to $\dot{\boldsymbol{\theta}}$ through the inverse Jacobian matrix $J^+(\boldsymbol{\theta})$ depends on the joint angle vector $\boldsymbol{\theta}$ at any given time.

Run InvJacobianMat.m to obtain the Jacobian and its pseudo inverse matrix.

$$
\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = J^+(\boldsymbol{\theta}) \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \end{bmatrix} \tag{1.60}
$$

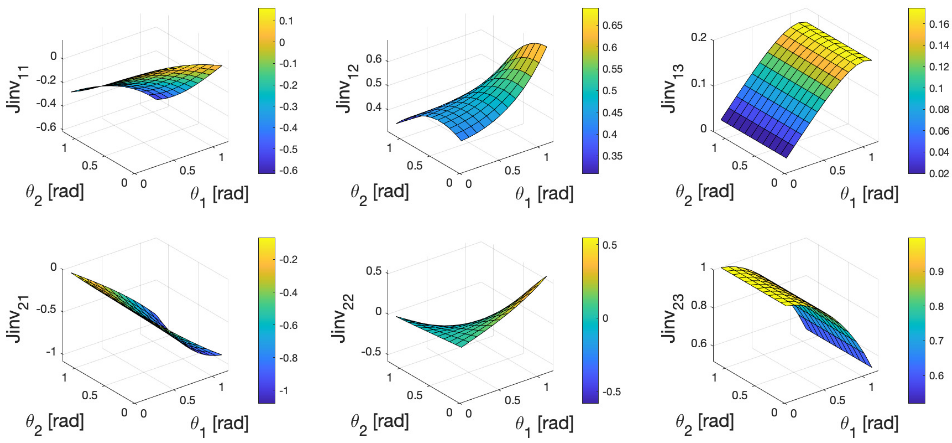


Figure 1.20: Elements of the $J^{+}(\boldsymbol{\theta})$ matrix for $\theta_1, \theta_2 \in [0,\ 1.2]\ [rad]$.

Chapter 2

Denavit-Hartenberg convention

2.1 Why does a design engineer need to know about this?

I am sure the above discussion has raised the question about guidelines to assign coordinate systems to objects. When different people assign coordinates to different objects in different ways, we need a lot of explanation about how each coordinate system is assigned. But if we have a convention, then we need little explanation every time we assign coordinates on a robot. Once we agree on a convention, we can define a set of parameters that design engineers can use to fill in a standard form of a homogeneous transformation matrix to avoid miscommunication among different design teams.

2.2 Intended learning outcomes

At the end of this lecture you should be able to

1. Know how to use Denavit-Hartenberg convention to transform vectors from one coordinate frame to another.
2. Know how to use symbolic Matlab codes to derive the compound homogeneous transformation matrix and the Jacobian matrix of a multi degree of freedom robot.
3. Know how to apply the knowledge to both rigid link and robots with constant curvature continuum links.

2.3 Frame assignment convention

Denavit and Hartenberg convention is one such widely followed conventions in robotics. One version is shown in figure 2.1. Link frame attachment procedure.

1. Identify the joint axes and draw lines along them.
2. Identify the common perpendiculars between them, or points of intersection.
3. At the point of intersection, or at the point where the common perpendicular meets the ith axis, assign the link frame origin.

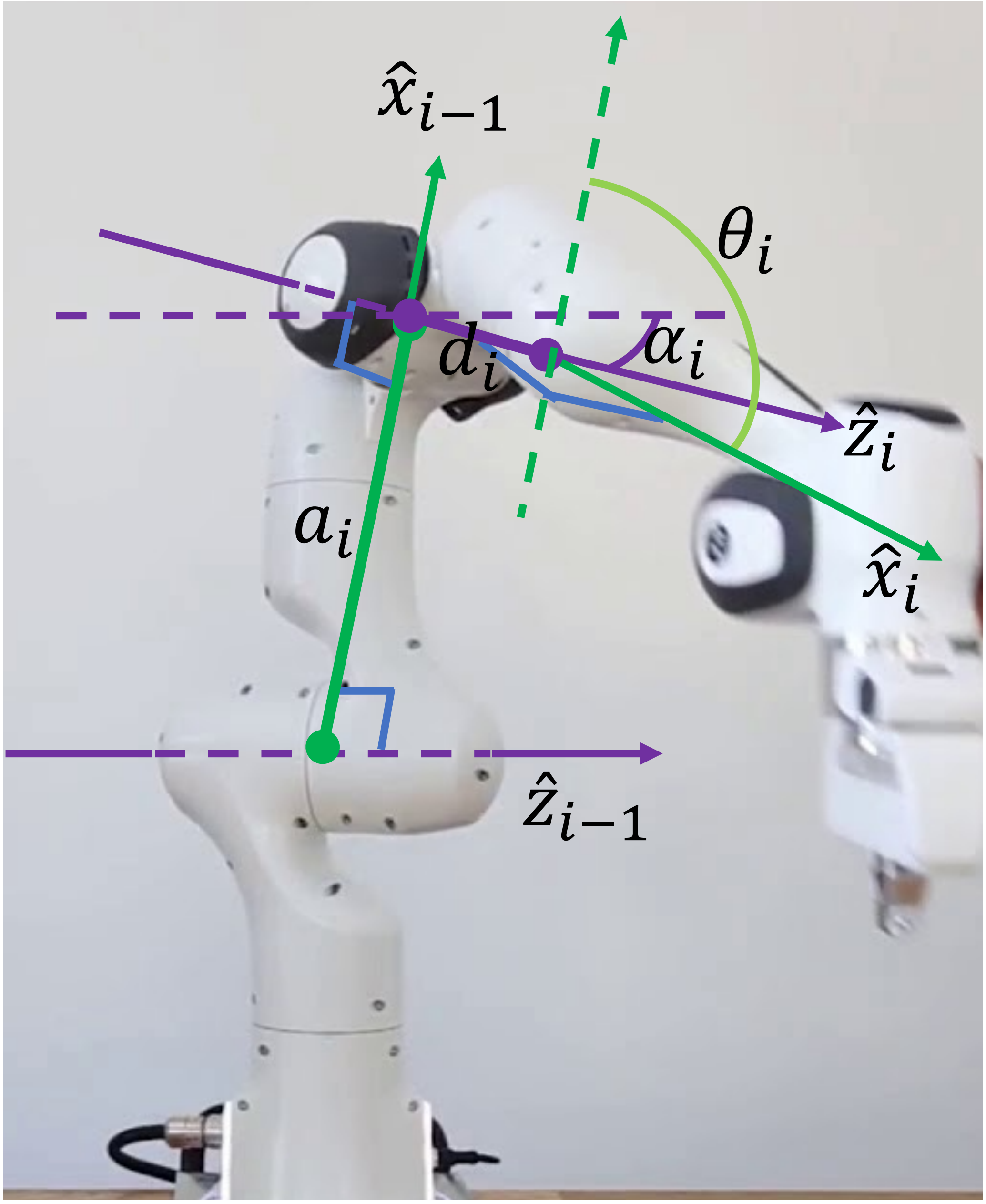


Figure 2.1: D-H parameters [2]. (source: [1], section 1.2). Four parameters are used to define the kinematics of each joint of a manipulator.

4. Assign the $\hat{\boldsymbol{z}}_i$ axis pointing along the ith joint axis following the right hand rule (how a screw moves).

5. Assign $\hat{\boldsymbol{x}}_i$ axis pointing along the common perpendicular, or if the $\hat{\boldsymbol{z}}_i$ and $\hat{\boldsymbol{z}}_{i+1}$ axes intersect, assign $\hat{\boldsymbol{x}}_i$ to the normal to the plane containing the two axes.

6. Assign the $\hat{\boldsymbol{y}}_i$ to complete the right-hand coordinate system (thumb points at $\hat{\boldsymbol{z}}_i$, index finger points at $\hat{\boldsymbol{x}}_i$, and the middle finger points at $\hat{\boldsymbol{y}}_i$).

The base frame $\{0\}$ doesn't move.

1. Since frame $\{0\}$ is arbitrary, it is simple to choose $\hat{\boldsymbol{z}}_0$ along the direction of axis-1.

2. Orient frame $\{0\}$ so that frame $\{1\}$ coincides with frame $\{0\}$ when the joint angle is zero.

The last frame $\{N\}$ is also arbitrary. Choose $\hat{\boldsymbol{x}}_N$ freely, but so as to cause as many linkage parameters as possible to zero.

In this module, we follow the convention given in Khalil and Dombre [1].

1. a_i = the distance from $\hat{\boldsymbol{z}}_{i-1}$ to $\hat{\boldsymbol{z}}_i$ measured along $\hat{\boldsymbol{x}}_{i-1}$.
2. α_i = the angle between $\hat{\boldsymbol{z}}_{i-1}$ and $\hat{\boldsymbol{z}}_i$ measured about $\hat{\boldsymbol{x}}_{i-1}$.
3. d_i = the distance from $\hat{\boldsymbol{x}}_{i-1}$ to $\hat{\boldsymbol{x}}_i$ measured along $\hat{\boldsymbol{z}}_i$.
4. θ_i = the angle between $\hat{\boldsymbol{x}}_{i-1}$ to $\hat{\boldsymbol{x}}_i$ measured about $\hat{\boldsymbol{z}}_i$.

Please watch this video for an animation of DH parameter assignment. Then,

$$ {}^{i-1}\boldsymbol{T}_i = \boldsymbol{R}_{\hat{x}_{i-1}}(\alpha_i)\ \boldsymbol{D}_{\hat{x}_{i-1}}(a_i)\ \boldsymbol{R}_{\hat{z}_i}(\theta_i)\ \boldsymbol{D}_{\hat{z}_i}(d_i) \tag{2.1} $$

$$ {}^{i-1}\boldsymbol{T}_i = \mathrm{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i)\ \mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i) \tag{2.2} $$

where ${}^{i-1}\boldsymbol{T}_i$ is read as frame i transformed to frame $i-1$. In lay language it means how frame i appears when it is viewed from frame $i-1$. The objective of this transformation is to view the movements of the robot from a base frame. This transformation can be done using two screws.

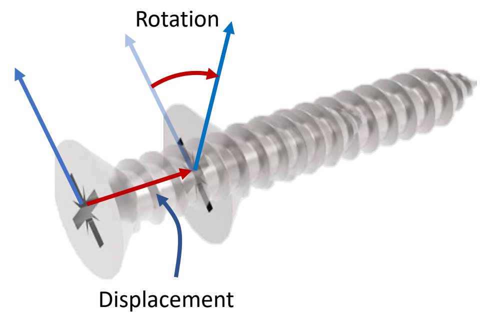


Figure 2.2: The notion of a screw in simultaneous rotation and translation.

Figure 2.2 shows the notion of a screw, where a simultaneous rotation and a translation takes place. For instance, $\mathrm{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i)$ can be seen as a screw that moves by a_i along the $\hat{x}_{i-1}$ axis when it rotates by α_i around $\hat{x}_{i-1}$. Therefore, the transformation from frame i to frame $i-1$ can be done in two steps. The first screw $\mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i)$ transforms frame i to an intermediate frame at origin O_{int} in figure 2.3. Then $\mathrm{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i)$ transforms the frame at origin O_{int} to the frame at origin O_{i-1}. Then the compound transformation ${}^{i-1}\boldsymbol{T}_i$ is given by $\mathrm{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i)\ \mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i)$ because $\mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i)$ was implemented first.

$$ \mathrm{Screw}_{\hat{z}_i}(d_i, \theta_i) = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.3} $$

Figure 2.3 shows the screw representing the displacement of d_i from the intermediate origin O_{int} to O_i along z_i with a corresponding rotation from x_{i-1} to x_i about z_i by an angle θ_i. We leave θ_i to be a generic positive variable according to the screw notation, so that the sign can be considered when we substitute numerical values.

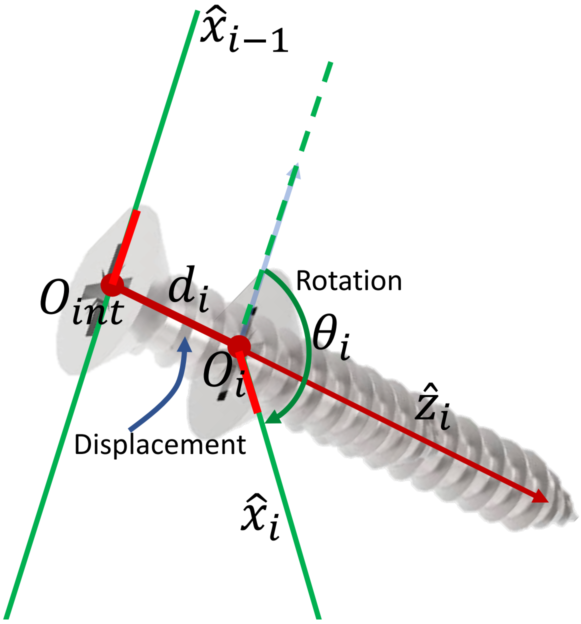


Figure 2.3: The screw representing the displacement of d_i from the intermediate origin O_{int} to O_i with a corresponding rotation from x_{i-1} to x_i by an angle θ_i.

$$\text{Screw}_{\hat{x}_{i-1}}(a_i, \alpha_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c\alpha_i & -s\alpha_i & 0 \\ 0 & s\alpha_i & c\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.4}$$

Figure 2.4 shows the screw representing the displacement of a_i from the origin O_{i-1} to the intermediate origin O_{int} along x_{i-1} with a corresponding rotation from z_{i-1} to z_i about x_{i-1} by an angle α_i. Please note that in this specific case, α_i is shown in the negative direction. However, we leave α_i to be a generic positive variable according to the screw notation, so that the sign can be considered when we substitute numerical values.

This leads to

$$^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.5}$$

It is very important to notice that at this point, we do not worry about the exact sign of the variables involved, because this is a generic case. We assign the sign in a specific case of a robot manipulator using the right hand rule also seen in the screw (a screw moves in when it is rotated clockwise). The homogeneous transformation matrix is in the form given by

$$^{i-1}\boldsymbol{T}_i = \begin{bmatrix} ^{i-1}\boldsymbol{R}_i & ^{i-1}\boldsymbol{p}_i \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \tag{2.6}$$

where

$$^{i-1}\boldsymbol{R}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i \end{bmatrix} \tag{2.7}$$

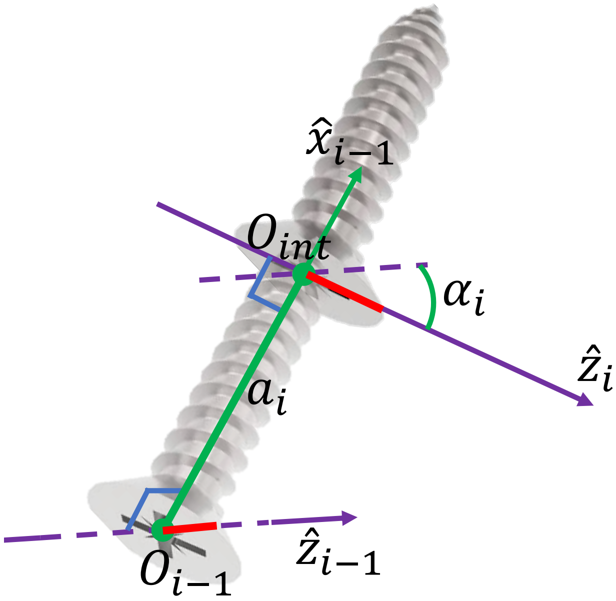


Figure 2.4: The screw representing the displacement of a_i from the origin O_{i-1} to the intermediate origin O_int with a corresponding rotation from z_{i-1} to z_i by an angle α_i.

and

$$^{i-1}\boldsymbol{p}_i = \begin{bmatrix} a_i \\ -s\alpha_i d_i \\ c\alpha_i d_i \end{bmatrix} \tag{2.8}$$

in Cartesian coordinates and

$$^{i-1}\boldsymbol{p}_i = \begin{bmatrix} a_i \\ -s\alpha_i d_i \\ c\alpha_i d_i \\ 1 \end{bmatrix} \tag{2.9}$$

in homogeneous coordinates.

Run DHDombreKhalil.m

2.4 Examples for computation of end effector position and orientation relative to the base frame

Example of a Scara robot manipulator

Fill the D-H table for the Scara robot manipulator shown in 2.5. We can draw the axes following the above guidelines and fill the DH table given in figure 2.6

1. Find the position of the tip of the manipulator relative to the base frame.

2. What is the orientation of the second link relative to the base frame?

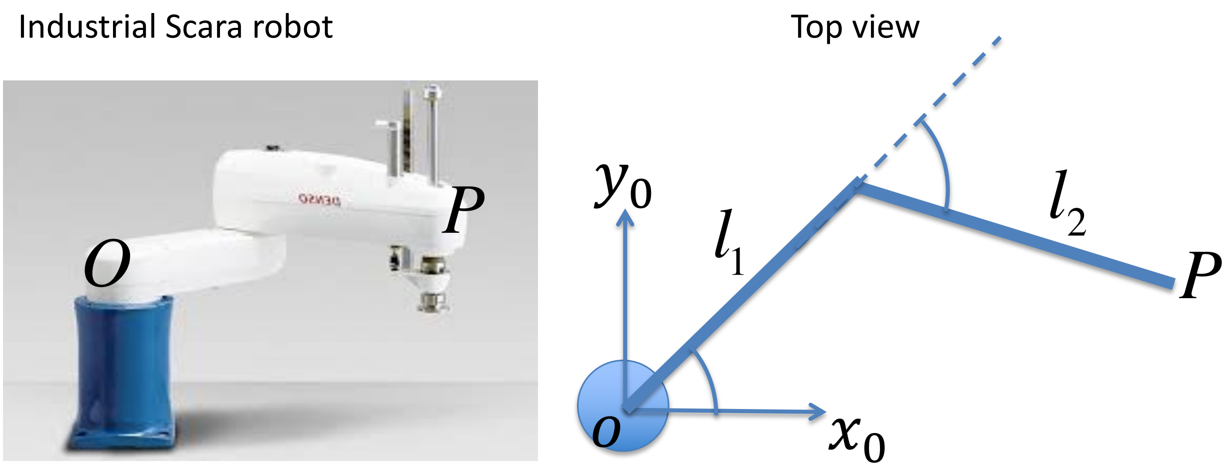


Figure 2.5: Scara robot manipulator

Using the DH table and equation (2.5) we can write:

$$
{}^0\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.10}
$$

$$
{}^1\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_1 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.11}
$$

$$
{}^2\boldsymbol{T}_3 = \begin{bmatrix} 1 & 0 & 0 & l_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.12}
$$

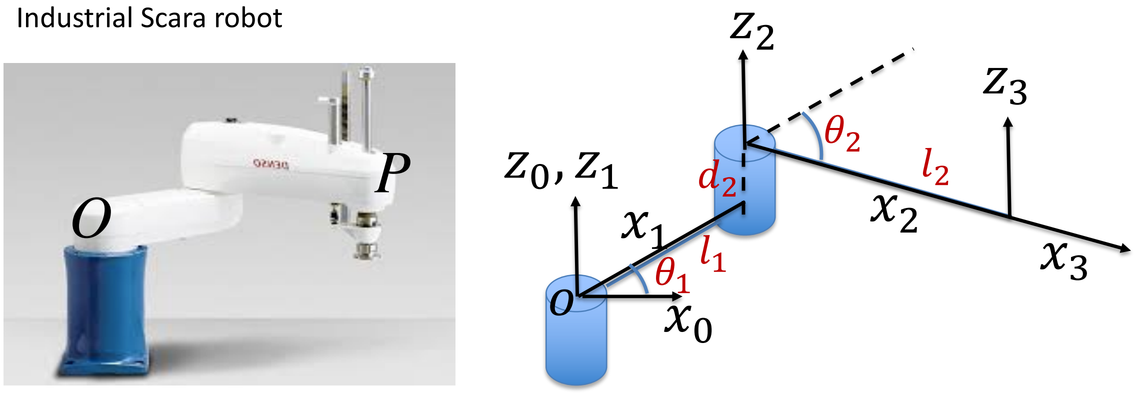
Run Scara.m to obtain the homogeneous transformation matrix
${}^0\boldsymbol{T}_3 =^0 \boldsymbol{T}_1\ {}^1\boldsymbol{T}_2\ {}^2\boldsymbol{T}_3$

$$
{}^0\boldsymbol{T}_3 = \begin{bmatrix} c(\theta_1+\theta_2) & -s(\theta_1+\theta_2) & 0 & l_1 c\theta_1 + l_2 c(\theta_1+\theta_2) \\ s(\theta_1+\theta_2) & c(\theta_1+\theta_2) & 0 & l_1 s\theta_1 + l_2 s(\theta_1+\theta_2) \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.13}
$$

Since,

$$
{}^j\boldsymbol{T}_i = \begin{bmatrix} {}^j\boldsymbol{R}_i & {}^j\boldsymbol{p}_i \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \tag{2.14}
$$

the position vector of the end-effector in the base frame (x_0, y_0, z_0) is given by the last column of ${}^0\boldsymbol{T}_3$:



i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	0	d_2	θ_2
3	l_2	0	0	0

Figure 2.6: The DH table for the Scara robot manipulator.

$$P = \begin{bmatrix} l_1 c\theta_1 + l_2 c(\theta_1 + \theta_2) \\ l_1 s\theta_1 + l_2 s(\theta_1 + \theta_2) \\ d_2 \\ 1 \end{bmatrix} \tag{2.15}$$

Very important: A visual inspection of figure 2.6 shows that θ_1 is positive and θ_2 is negative in this particular configuration. However, DH-parameters give a generic equation of the position vector of the end-effector in the base frame (x_0, y_0, z_0). In equation 2.15 you will notice that there are terms $c(\theta_1 + \theta_2)$ and $s(\theta_1 + \theta_2)$. These mean that you have to substitute the exact values of θ_1 and θ_2 with their signs to obtain the numerical values of the elements in the position vector.

To obtain the orientation between different axes, we use our knowledge about the orientation matrix obtained using projections between unit vectors:

$${}^0\boldsymbol{R}_3 = \begin{bmatrix} \hat{\boldsymbol{x}}_3 \cdot \hat{\boldsymbol{x}}_0 & \hat{\boldsymbol{y}}_3 \cdot \hat{\boldsymbol{x}}_0 & \hat{\boldsymbol{z}}_3 \cdot \hat{\boldsymbol{x}}_0 \\ \hat{\boldsymbol{x}}_3 \cdot \hat{\boldsymbol{y}}_0 & \hat{\boldsymbol{y}}_3 \cdot \hat{\boldsymbol{y}}_0 & \hat{\boldsymbol{z}}_3 \cdot \hat{\boldsymbol{y}}_0 \\ \hat{\boldsymbol{x}}_3 \cdot \hat{\boldsymbol{z}}_0 & \hat{\boldsymbol{y}}_3 \cdot \hat{\boldsymbol{z}}_0 & \hat{\boldsymbol{z}}_3 \cdot \hat{\boldsymbol{z}}_0 \end{bmatrix} \tag{2.16}$$

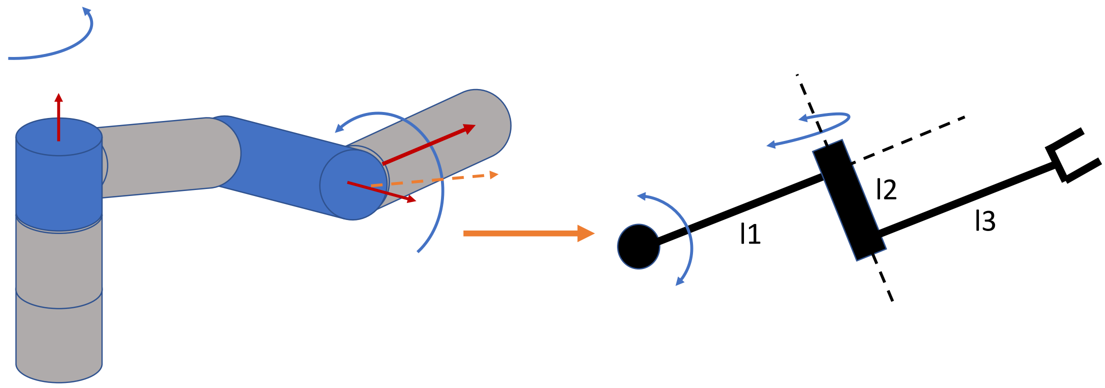
$${}^0\boldsymbol{R}_3 = \begin{bmatrix} c(\theta_1 + \theta_2) & -s(\theta_1 + \theta_2) & 0 \\ s(\theta_1 + \theta_2) & c(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2.17}$$

The orientation of the last link relative to the base frame is given by $\arccos(\hat{x}_3 \cdot \hat{x}_0) = (\theta_1 + \theta_2)$

This orientation is given as a generic equation. You have to substitute the exact values of θ_1 and θ_2 with their signs to obtain the numerical value of orientation. Similarly check if the angles between other pairs of axes make sense.

Example for manipulators with perpendicular axes

Assign the frames following the D-H convention of Khalil and Dombre for the following manipulator. Hence find the position of the end point relative to the top of the first vertical link.



1. a_i = the distance from $\hat{z}_{i-1}$ to $\hat{z}_i$ measured along $\hat{x}_{i-1}$.
2. α_i = the angle between $\hat{z}_{i-1}$ and $\hat{z}_i$ measured about $\hat{x}_{i-1}$.
3. d_i = the distance from $\hat{x}_{i-1}$ to $\hat{x}_i$ measured along $\hat{z}_i$.
4. θ_i = the angle between $\hat{x}_{i-1}$ to $\hat{x}_i$ measured about $\hat{z}_i$.

Then we use the frame assignment shown in figure 2.8 to fill the DH table.

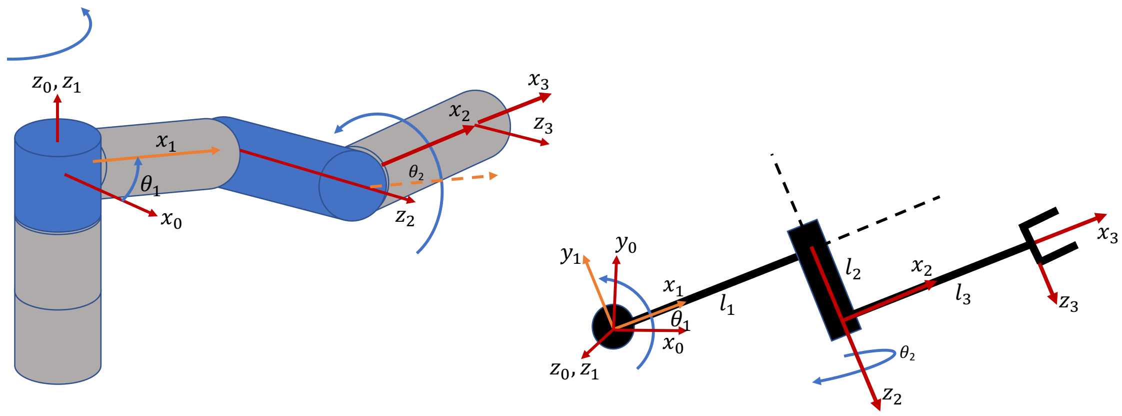


Figure 2.7: Alternative-1: x_2 axis along the second link.

Then the D-H table is given by

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	90	l_2	θ_2
3	l_3	0	0	0

$$ {}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.18} $$

Then we can use the DH table and equation (2.23) to write down the transformation matrices given by,

$$ {}^0\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.19} $$

$$ {}^1\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_1 \\ 0 & 0 & -1 & -l_2 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.20} $$

$$ {}^2\boldsymbol{T}_3 = \begin{bmatrix} 1 & 0 & 0 & l_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.21} $$

${}^0\boldsymbol{T}_3 = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2\,{}^2\boldsymbol{T}_3$

$$ {}^0\boldsymbol{T}_3 = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 & l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 & l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ s\theta_2 & c\theta_2 & 0 & l_3 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.22} $$

Run ExamplewithPerpAxes.m Alternative-1.

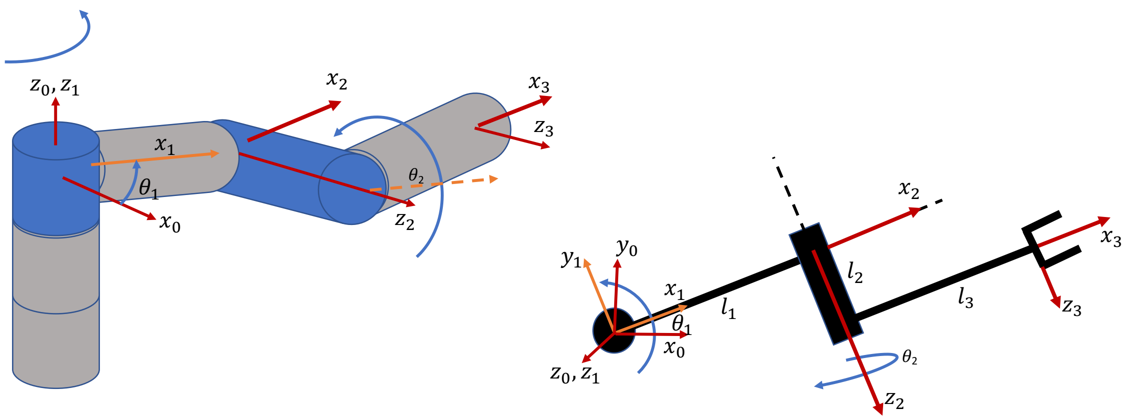


Figure 2.8: Alternative-2: x_2 axis in the same plane of x_1.

Then the D-H table is given by

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	l_1	90	0	θ_2
3	l_3	0	l_2	0

$$ {}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.23} $$

Then we can use the DH table and equation (2.23) to write down the transformation matrices given by,

$$ {}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.24} $$

$$ {}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_1 \\ 0 & 0 & -1 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.25} $$

$$ {}^{2}\boldsymbol{T}_3 = \begin{bmatrix} 1 & 0 & 0 & l_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.26} $$

${}^{0}\boldsymbol{T}_3 ={}^{0} \boldsymbol{T}_1 \, {}^{1}\boldsymbol{T}_2 \, {}^{2}\boldsymbol{T}_3$

$$ {}^{0}\boldsymbol{T}_3 = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 & l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 & l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ s\theta_2 & c\theta_2 & 0 & l_3 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.27} $$

Run ExamplewithPerpAxes.m Alternative-2.
since

$$ {}^{j}\boldsymbol{T}_i = \begin{bmatrix} {}^{j}\boldsymbol{R}_i & {}^{j}\boldsymbol{p}_i \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \tag{2.28} $$

the position of the end point relative to the top of the first vertical link is given by

$$ {}^{0}\boldsymbol{P}_3 = \begin{bmatrix} l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ l_3 s\theta_2 \end{bmatrix} \tag{2.29} $$

Again, please note that DH-parameters give a generic equation of the position vector of the end-effector in the base frame (x_0, y_0, z_0). You have to substitute the exact values of θ_1 and θ_2 with their signs to obtain the numerical values of the elements in the position vector.

2.5 Solving the 3D inverse kinematics problem using the Jacobian matrix

2.5.1 Mapping joint velocities to end point Cartesian velocities

Once we know the homogeneous transformation matrix for a robot manipulator, we can obtain the rotation matrix which gives the orientation of the last link relative to the base link, and the position of the end effector relative to the base coordinate system.
A matrix known as the Jacobian matrix can be derived from the position vector to solve the inverse kinematics problem.
Let the position vector of the end effector of a robot manipulator be given by $\boldsymbol{p} = f(\boldsymbol{\theta})$, where

$\boldsymbol{\theta}$ is the joint angle vector. Then we can use the chain rule to obtain the speed vector of the end effector with respective to the joint speeds given by

$$\frac{d\boldsymbol{p}}{dt} = \frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \frac{d\boldsymbol{\theta}}{dt} \tag{2.30}$$

Then the Jacobian matrix $J(\boldsymbol{\theta}) = \frac{\partial f(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$.

$$\dot{\boldsymbol{p}} = J(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \tag{2.31}$$

Then, if J is a square matrix, we can obtain the joint angular speeds given an end-point speed vector using $\dot{\boldsymbol{\theta}} = J(\boldsymbol{\theta})^{-1}\dot{\boldsymbol{p}}$. If this can be computed, a given desired end-effector position profile can first be differentiated to obtain the end-effector speed profile, and then the above relationship can be used to obtain the joint angular speed profile. Then the joint angle profile can be obtained by integrating the joint angular speed profile.

However, in most cases, the Jacobian matrix is not a square matrix due to more than 3-joints in the manipulator. In such cases, we use the Moore-Penrose pseudo-inverse of the Jacobian given by $J^{+} = J^{T}(JJ^{T})^{-1}$.

Please note that inverse kinematics solved using this approach will give a joint angle velocity profile depending on the initial joint angle configuration $\boldsymbol{\theta}_{t=0}$ fed to $J(\boldsymbol{\theta})^{-1}$. This can give multiple joint space solutions for redundant robots (where $J(\boldsymbol{\theta})$ is not a square matrix, or in other words, $J(\boldsymbol{\theta})$ is rank deficient) depending on the initial joint configuration.

2.5.2 Mapping external forces to joint torques

Here we use the principle of virtual work. In virtual work, the total work done by moving parts of a system is zero. In the case of a robotic manipulator, the work done by the torques around joints for a small joint movement is equal to the work done by the force vectors at contact points on the displacements of those points.

Virtual work done by the torque vector $\boldsymbol{\tau}$ on a joint vector $\boldsymbol{\theta}$ = virtual work done by the force vector $\boldsymbol{f}$ on a displacement vector $\boldsymbol{p}$.

$$\begin{aligned}
\boldsymbol{\tau}^T \delta\boldsymbol{\theta} &= \boldsymbol{f}^T \delta\boldsymbol{p} \qquad (2.32)\\
\lim_{\delta t \to 0} \boldsymbol{\tau}^T \frac{\delta\boldsymbol{\theta}}{\delta t} &= \lim_{\delta t \to 0} \boldsymbol{f}^T \frac{\delta\boldsymbol{p}}{\delta t}\\
\boldsymbol{\tau}^T \dot{\boldsymbol{\theta}} &= \boldsymbol{f}^T \dot{\boldsymbol{p}}\\
\boldsymbol{\tau}^T \dot{\boldsymbol{\theta}} &= \boldsymbol{f}^T J \dot{\boldsymbol{\theta}}\\
\boldsymbol{\tau}^T &= \boldsymbol{f}^T J\\
\boldsymbol{\tau} &= J^T \boldsymbol{f}
\end{aligned}$$

(2.33)

Example for the robot manipulator given in figure 2.8

We obtained that the position vector is given by

$${}^{0}\boldsymbol{P}_3 = \begin{bmatrix} l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ l_3 s\theta_2 \end{bmatrix} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \tag{2.34}$$

Then the Jacobian is given by

$$J(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial p_x}{\partial \theta_1} & \frac{\partial p_x}{\partial \theta_2} \\ \frac{\partial p_y}{\partial \theta_1} & \frac{\partial p_y}{\partial \theta_2} \\ \frac{\partial p_z}{\partial \theta_1} & \frac{\partial p_z}{\partial \theta_2} \end{bmatrix} \tag{2.35}$$

Run JacobianMat.m to obtain the Jacobian matrix.

$$J(\boldsymbol{\theta}) = \begin{bmatrix} l_2\cos(\theta_1) - l_1\sin(\theta_1) - l_3\cos(\theta_2)\sin(\theta_1), & -l_3\cos(\theta_1)\sin(\theta_2) \\ l_1\cos(\theta_1) + l_2\sin(\theta_1) + l_3\cos(\theta_1)\cos(\theta_2), & -l_3\sin(\theta_1)\sin(\theta_2) \\ 0, & l_3\cos(\theta_2) \end{bmatrix} \tag{2.36}$$

Now you can see that $J(\boldsymbol{\theta})$ has elements that are functions of $\boldsymbol{\theta}$. Because of this reason the map from end point velocities to joint space velocities through the Jacobian matrix depends on the joint angle vector at any given time. This matters if the robot moves to joint configurations such as singular postures where some elements in the inverse/pseudo-inverse of the Jacobian matrix can grow to be much larger than others. As an exercise, compute the pseudo-inverse of the above Jacobian matrix and predict what joint angles would cause some elements to grow much larger than the others.

Run InvJacobianMat.m to obtain the Jacobian matrix. You will notice that the symbolic equation is very long. Therefore, I am not going to type it here, but evaluate its elements for different joint angle values and use plots to make sense of the behavior of the Pseudo-Inverse of the Jacobian for different joint angles.

Figure 2.9 shows how each element in the pseudoinverse J^+, for $\theta_1, \theta_2 \in [-\pi, \pi]$. Imagine, the robot manipulator is in $\theta_1 = \theta_2 \approx -2$ [rad]. Then $(1,1)$ element and $(1,3)$ element will have relatively large values compared to other elements. This would make $\dot{p}_x$ and $\dot{p}_z$ have a disproportionately higher influence on $\dot{\theta}_1$ while slowing down joint-2 with lower values for $\dot{\theta}_2$.

Please modify InvJacobianMat.m to study this in more detail till you get a deep insight into this very important idea used in most industry applications.

2.6 Some biomechanics examples

Robotic Hoof:

Assign the frames following the D-H convention of Khalil and Dombre for the robotic Hoof. Hence find its homogeneous transformation matrix from the fetlock joint to the tip of the claw. Derive the Jacobian matrix that maps joint velocities to Cartesian velocities of the tip.

$${}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.37}$$

$${}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & -1 & L_1 + L_2 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.38}$$

$${}^{2}\boldsymbol{T}_3 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.39}$$

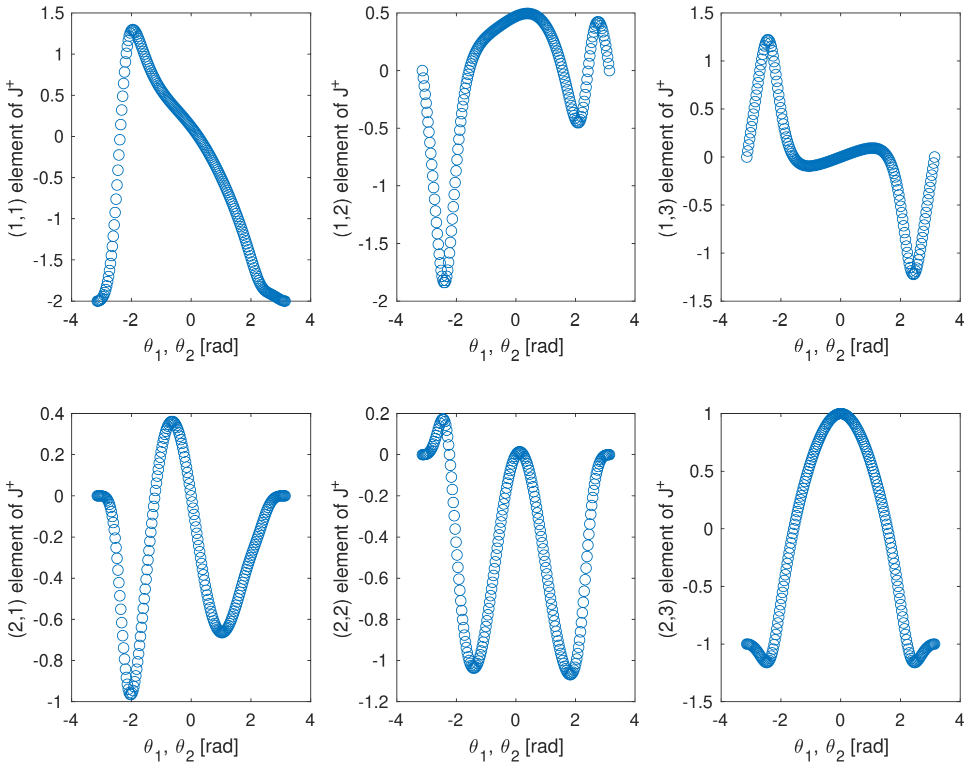
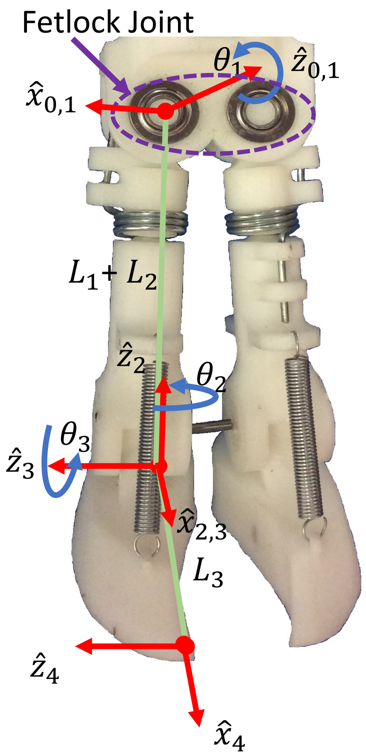


Figure 2.9: The behavior of each element in the pseudoinverse J^+, for $\theta_1, \theta_2 \in [-\pi, \pi]$.



i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	0	90	$-L_1 - L_2$	θ_2
3	0	90	0	θ_3
4	L_3	0	0	0

$$
{}^{3}\boldsymbol{T}_4 = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.40}
$$

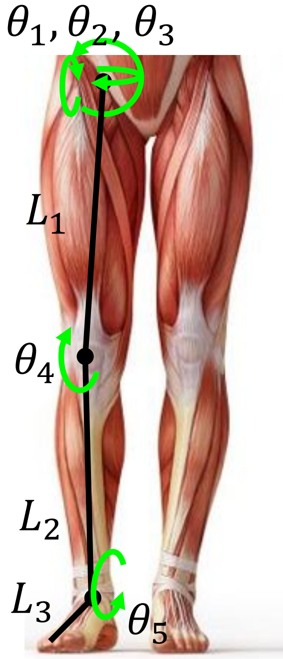
$$
{}^0\boldsymbol{T}_4 = {}^0\boldsymbol{T}_1\,{}^1\boldsymbol{T}_2\,{}^2\boldsymbol{T}_3\,{}^3\boldsymbol{T}_4
$$

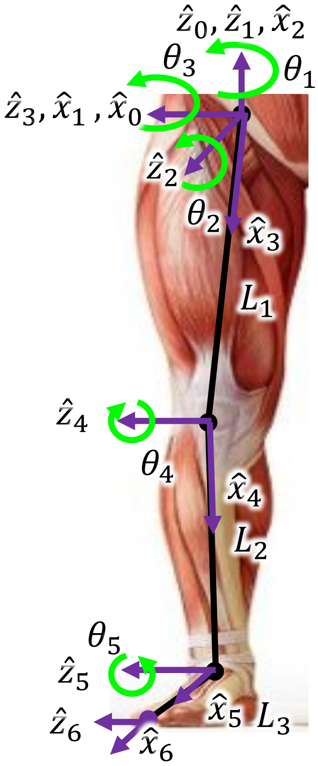
$$
\begin{bmatrix}
c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3 & c\theta_3 s\theta_1 - c\theta_1 c\theta_2 s\theta_3 & c\theta_1 s\theta_2 & L_3(c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3) - (L_1 + L_2)s\theta_1 \\
-c\theta_1 s\theta_3 + c\theta_2 c\theta_3 s\theta_1 & -c\theta_1 c\theta_3 - c\theta_2 s\theta_1 s\theta_3 & s\theta_1 s\theta_2 & -L_3(c\theta_1 s\theta_3 - s\theta_1 c\theta_2 c\theta_3) + (L_1 + L_2)c\theta_1 \\
c\theta_3 s\theta_2 & -s\theta_2 s\theta_3 & -c\theta_2 & L_3 c\theta_3 s\theta_2 \\
0 & 0 & 0 & 1
\end{bmatrix} \tag{2.41}
$$

$$
\boldsymbol{p} = \begin{bmatrix}
L_3(c\theta_1 c\theta_2 c\theta_3 + s\theta_1 s\theta_3) - (L_1 + L_2)s\theta_1 \\
-L_3(c\theta_1 s\theta_3 - s\theta_1 c\theta_2 c\theta_3) + (L_1 + L_2)c\theta_1 \\
L_3 c\theta_3 s\theta_2
\end{bmatrix} \tag{2.42}
$$

$$
J = \begin{bmatrix}
L_3(c\theta_1 s\theta_3 - c\theta_2 c\theta_3 s\theta_1) - c\theta_1(L_1 + L_2) & -L_3 c\theta_1 c\theta_3 s\theta_2 & L_3(c\theta_3 s\theta_1 - c\theta_1 c\theta_2 s\theta_3) \\
L_3(s\theta_1 s\theta_3 + c\theta_1 c\theta_2 c\theta_3) - s\theta_1(L_1 + L_2) & -L_3 c\theta_3 s\theta_1 s\theta_2 & -L_3(c\theta_1 c\theta_3 + c\theta_2 s\theta_1 s\theta_3) \\
0 & L_3 c\theta_2 c\theta_3 & -L_3 s\theta_2 s\theta_3
\end{bmatrix} \tag{2.43}
$$

Human Leg: Assign the frames following the D-H convention for the human leg and find its homogeneous transformation matrix.





i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	0	-90	0	$-\theta_2$
3	0	-90	0	$-\theta_3$
4	L_1	0	0	θ_4
5	L_2	0	0	θ_5
6	L_3	0	0	0

The thumb and the foldable palm: Figure 2.10 shows an approximated kinematic model for the human thumb and the foldable palm. Find the D-H parameters for this model and use it to write a symbolic Matlab program to derive an equation for the position of joint J6 and the thumbtip. Substitute zero to all joint angles to check if your result makes sense.

Assume that $l_1 = 25$mm, $l_2 = 25$mm, $l_3 = 32$mm, $l_4 = 20$mm, $l_5 = 30$mm, $l_6 = 20$mm, and $\gamma = 60$ degrees.

Derive the Jacobian matrix that maps joint velocities to Cartesian velocities of joint J6 and the thumbtip. The corresponding D-H table is given by

i	$a_i[m]$	$\alpha_i[deg]$	$d_i[m]$	$\theta_i[deg]$
$0'$	0	0	0	$180 + \gamma$
1	l_1	90	l_2	$90 + \theta_1$
2	0	90	l_3	θ_2
3	0	90	0	$90 + \theta_3$
4	0	90	0	$270 + \theta_4$
5	l_4	90	0	θ_5
6	0	-90	0	θ_6
7	l_5	90	0	θ_7
8	l_6	0	0	0

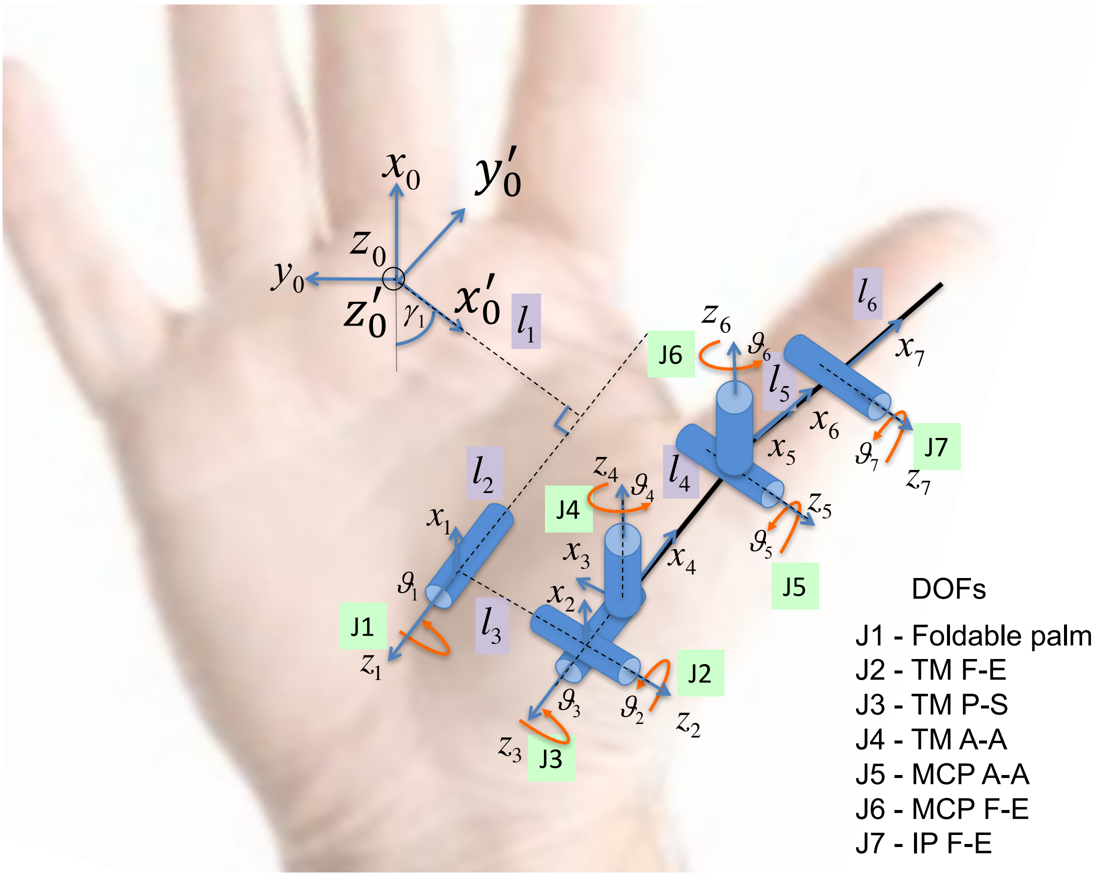


Figure 2.10: An approximated model for the thumb and the foldable palm

2.7 Continuum robots

Continuum robots are inspired by biological inspirations such as the elephant trunk, octopus arms, and chameleon tongue. They are still under research for applications such as minimally invasive surgery, malleable robotic arms, and fingers for soft object manipulation. Figure 2.11 shows an example we have used to palpate soft tissue [4]. However, modeling the kinematics and dynamics of soft continuum robots have not been as straightforward as rigid link robots. Here, we will have a brief look at some simplified models because there is a rapid growth in research and applications of soft robots in the recent past.

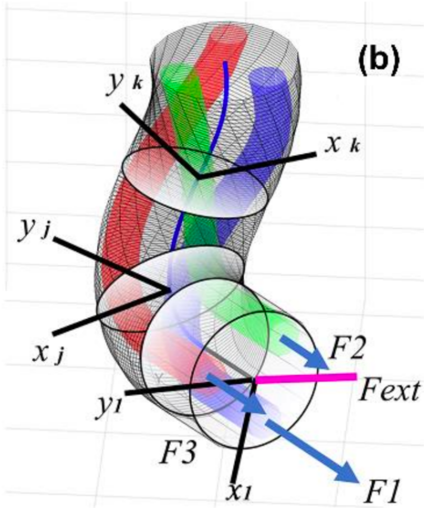
Soft robots are usually made out of different types of rubber. We use ecoflex in our laboratories to make soft robots. They can be controlled using a variety of actuators such as motor driven tendons, shape memory alloys, pneumatic or hydraulic pressure channels, or hydrogels.

Let us consider one segment of the soft continuum robot as shown in figure 2.13. We can make a simplifying assumption that the segment has a constant curvature of κ. Then the radius of the first segment is $r_1 = \frac{1}{\kappa_1}$, and the length of the central axis is s_1.

The frame assignment is shown in figure 2.12. Figure 2.13 shows the curvature parameters used to build the D-H table and to compute α and length l_1.

$\alpha = \frac{s_1}{r_1} = s_1\kappa_1$.

In this model, the base of the segment has 2 degrees of freedom to rotate around the z_0 axis and then to rotate horizontally around the z_2 axis. There is zero distance between these two axes of rotations. We get x_1, z_1 after rotating the base by θ_1 around z_0.



Sadati, S. H., Shiva, A., Herzig, N., Rucker, C. D., Hauser, H., Walker, I. D., ... & Nanayakkara, T. (2020). Stiffness Imaging With a Continuum Appendage: Real-Time Shape and Tip Force Estimation From Base Load Readings. *IEEE Robotics and Automation Letters*, *5*(2), 2824-2831.

Figure 2.11: A continuum segment driven by hydraulic pressure channels

The end of the segment has a coordinate frame with the z_3 axis parallel to z_2. There is zero rotation around this axis. But the advantage of having this frame is that we can easily define a common perpendicular x_2 between z_2 and z_3. Then, similar to the base, the next segment rotates around the z_4 and z_5 axes. For simplicity, we stop at the z_4 axis in this analysis to obtain the position of the tip of the first segment relative to the base.

There are three main kinematic parameters used to define the shape and orientation of a continuum segment in the Euclidean space:

- κ: Curvature of the segment (see figure 2.13).
- s: The length of the central axis of the segment (see figure 2.13).
- ϕ: The rotation of the plane of curvature from the x-axis (see figure 2.14).

Then the D-H table is given by

i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	0	90	0	θ_2
3	l_1	0	0	$-\left(90+\frac{\alpha}{2}\right)$
4	0	-90	0	θ_4

where,

$$\alpha = s_1\kappa_1$$

$$\begin{aligned} l_1 &= 2r_1 \sin\frac{\alpha}{2} \\ &= \frac{2}{\kappa_1}\sin\frac{s_1\kappa_1}{2} \end{aligned} \tag{2.44}$$

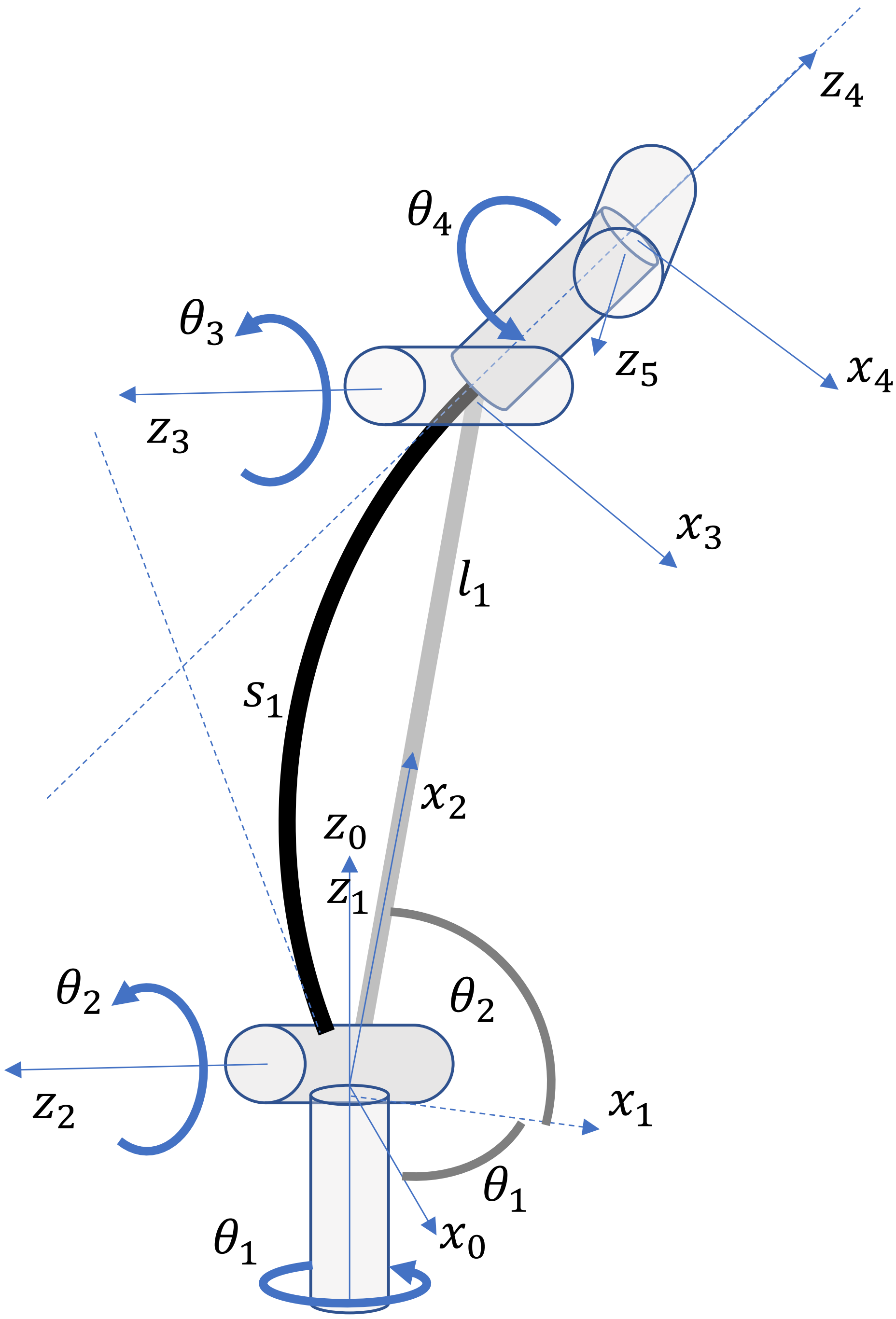


Figure 2.12: Frame assignment to a segment of a continuum robot with constant curvature assumption.

Run Continuumseg.m.

Let the length of channels (it can be tendons, hydraulic, or pneumatic channels) be given by l_r, l_g, and l_b as shown in figure 2.14. Then we can establish a relationship between the above kinematic parameters and the channel lengths (tendon lengths or pressure channels lengths) given by

$\alpha = \frac{l_c}{r} = l_c \kappa$

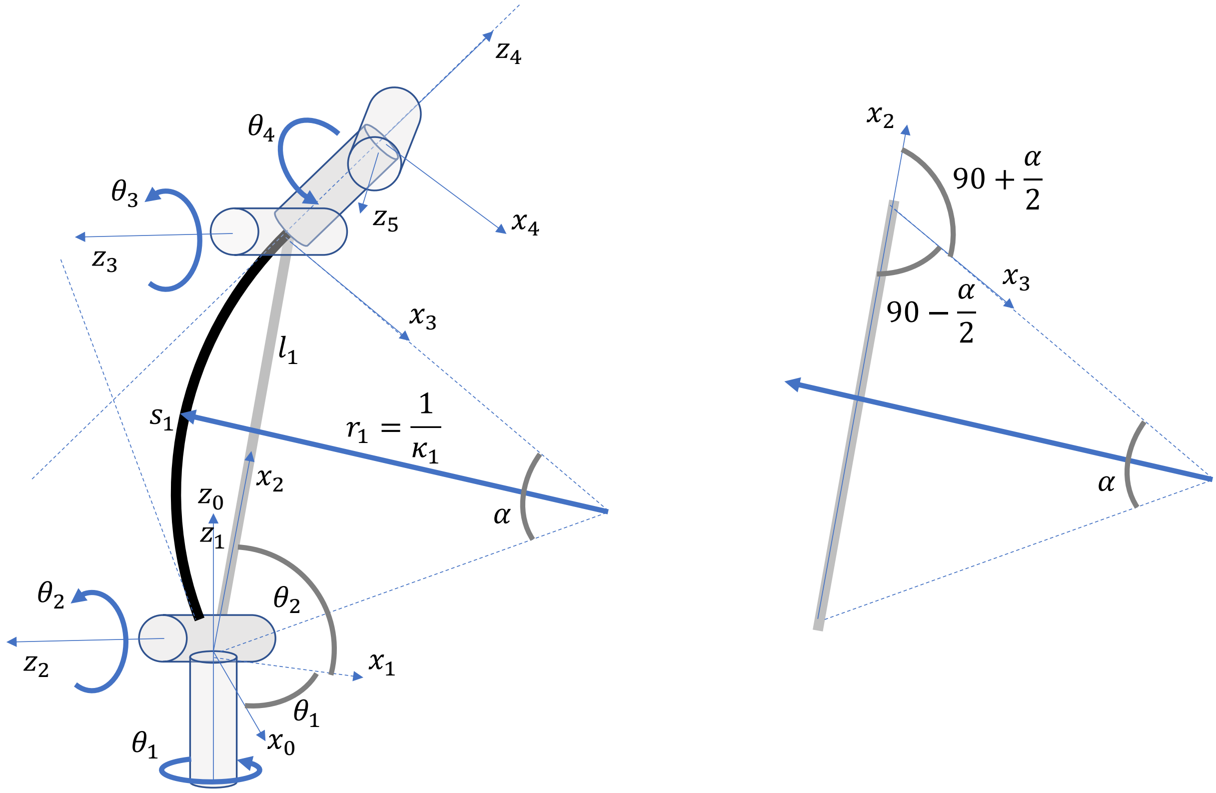


Figure 2.13: DH parameters for a soft continuum robot segment with constant curvature assumption.

$$l_c = \frac{l_r + l_b + l_g}{3}$$

where l_c is the length of the centre line in figure 2.14.

$$\phi = \tan^{-1}\left(\frac{l_g + l_b - 2l_r}{\sqrt{3}\,(l_g - l_b)}\right)$$

where ϕ is the angle of the plane passing through the centerline arch from the x axis as shown in figure 2.14.

$$\kappa = 2\frac{\sqrt{l_r^2 + l_g^2 + l_b^2 - l_r l_g - l_g l_b - l_r l_b}}{r_i\,(l_r + l_g + l_b)}$$

where κ is the curvature of the centerline arch as shown in figure figure 2.14.

For a detailed derivation, please refer to [3].

Bibliography

[1] Etienne Dombre and Wisama Khalil. *Modeling, performance analysis and control of robot manipulators*. Wiley Online Library, 2007.

[2] Richard S Hartenberg and Jacques Denavit. A kinematic notation for lower pair mechanisms based on matrices. 1955.

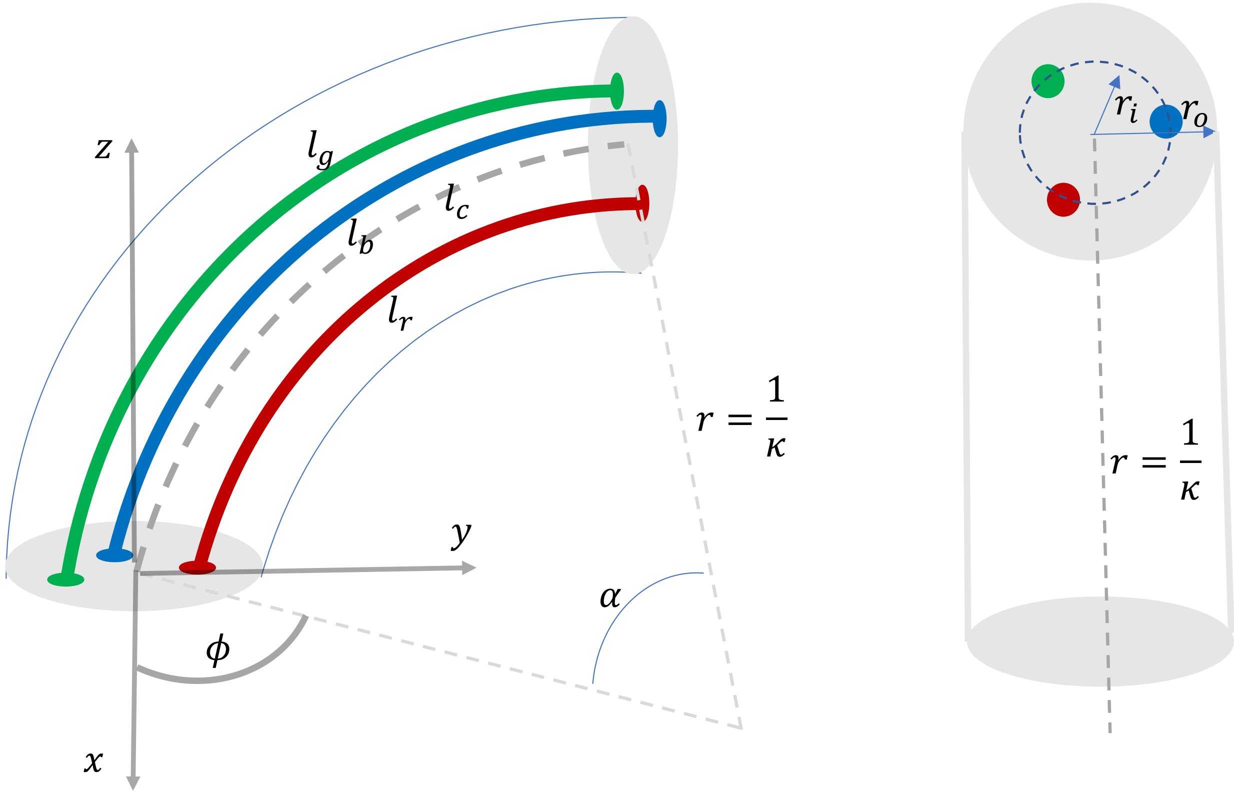


Figure 2.14: A continuum segment driven by tendons.

[3] Bryan A Jones and Ian D Walker. Kinematics for multisection continuum robots. *IEEE Transactions on Robotics*, 22(1):43–55, 2006.

[4] SM Hadi Sadati, Ali Shiva, Nicolas Herzig, Caleb D Rucker, Helmut Hauser, Ian D Walker, Christos Bergeles, Kaspar Althoefer, and Thrishantha Nanayakkara. Stiffness imaging with a continuum appendage: Real-time shape and tip force estimation from base load readings. *IEEE Robotics and Automation Letters*, 5(2):2824–2831, 2020.

Chapter 3

Past papers

3.1 2016 past paper: Q2. Angular and end-point speeds

This question will let you determine positions and orientations of a 3-link manipulator.

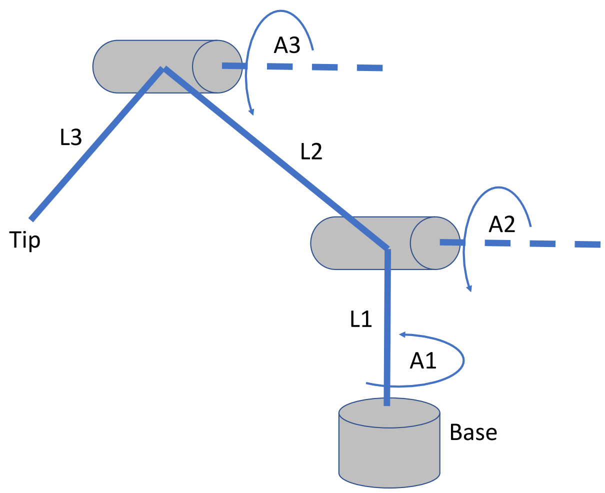


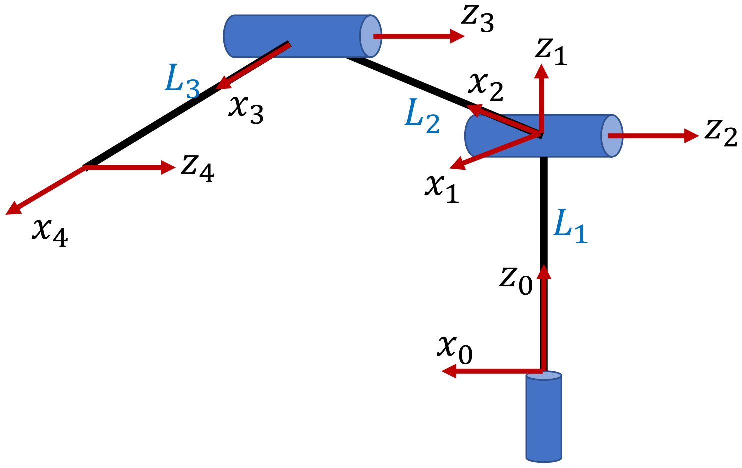
Figure 3.1: A 3-link robot manipulator with three revolute joints. The motors rotate around A_1-A_3 axises. The three links are L_1, L_2 and L_3 long, respectively.

a)

Use Denavit-Hartenberg (DH) convention to assign frames to each joint and fill up the DH table.

b)

Use the DH table in your answer to (a) above and the homogeneous transformation matrix in equation 3.10 to compute the position of the tip of the manipulator when $L_1 = 0.5$m, $L_2 = 0.25$m, $L_3 = 1$m, $\theta_1 = \pi/4$ rad, $\theta_2 = -\pi/3$ rad, and $\theta_3 = \pi/2$ rad. Please note: θ_i, $i = 1, 2, 3$ are joint angles, d_i is the distance from $\hat{\boldsymbol{x}}_{i-1}$ to $\hat{\boldsymbol{x}}_i$ measured along $\hat{\boldsymbol{z}}_i$, α_i is the angle between $\hat{\boldsymbol{z}}_{i-1}$ and $\hat{\boldsymbol{z}}_i$ measured about $\hat{\boldsymbol{x}}_{i-1}$, and a_i is the distance from $\hat{\boldsymbol{z}}_{i-1}$ to $\hat{\boldsymbol{z}}_i$ measured along $\hat{\boldsymbol{x}}_{i-1}$.



i	a_i	α_i	d_i	θ_i
1	0	0	L_1	θ_1
2	0	-90	0	θ_2
3	L_2	0	0	θ_3
4	L_3	0	0	0

Figure 3.2: The suggest ed D-H frame assignment

This is a special case. When there is a "T" junction like the first and the second joints from the base of the robot shown in figure 3.2, it is important to keep in mind the first frame should be assigned in the second horizontal motor with the base frame at the vertical axis. This allows to account for the first link (length L_1) as an offset parameter between the base and the first frame. If the two orthogonal motors are at zero distance apart, then the origins of these two frames coincide.

1. a_i = the distance from $\hat{\boldsymbol{z}}_{i-1}$ to $\hat{\boldsymbol{z}}_i$ measured along $\hat{\boldsymbol{x}}_{i-1}$.

2. α_i = the angle between $\hat{\boldsymbol{z}}_{i-1}$ and $\hat{\boldsymbol{z}}_i$ measured about $\hat{\boldsymbol{x}}_{i-1}$.

3. d_i = the distance from $\hat{\boldsymbol{x}}_{i-1}$ to $\hat{\boldsymbol{x}}_i$ measured along $\hat{\boldsymbol{z}}_i$.

4. θ_i = the angle between $\hat{\boldsymbol{x}}_{i-1}$ to $\hat{\boldsymbol{x}}_i$ measured about $\hat{\boldsymbol{z}}_i$.

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.1}
$$

$$
{}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.2}
$$

$$
{}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.3}
$$

$$
{}^{2}\boldsymbol{T}_3 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & L_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.4}
$$

$$
{}^{3}\boldsymbol{T}_4 = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.5}
$$

$$
{}^{0}\boldsymbol{T}_4 = {}^{0}\boldsymbol{T}_1\ {}^{1}\boldsymbol{T}_2\ {}^{2}\boldsymbol{T}_3\ {}^{3}\boldsymbol{T}_4
$$

$$
\begin{pmatrix} \cos(\theta_2+\theta_3)\ \cos(\theta_1) & -\sin(\theta_2+\theta_3)\ \cos(\theta_1) & -\sin(\theta_1) & \cos(\theta_1)\ (l_3\ \cos(\theta_2+\theta_3) + l_2\ \cos(\theta_2)) \\ \cos(\theta_2+\theta_3)\ \sin(\theta_1) & -\sin(\theta_2+\theta_3)\ \sin(\theta_1) & \cos(\theta_1) & \sin(\theta_1)\ (l_3\ \cos(\theta_2+\theta_3) + l_2\ \cos(\theta_2)) \\ -\sin(\theta_2+\theta_3) & -\cos(\theta_2+\theta_3) & 0 & l_1 - l_3\ \sin(\theta_2+\theta_3) - l_2\ \sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{3.6}
$$

$$
{}^{0}\boldsymbol{T}_4 = \begin{bmatrix} 0.6124 & -0.3536 & -0.7071 & 0.7008 \\ 0.6124 & -0.3536 & 0.7071 & 0.7008 \\ -0.5000 & -0.8660 & 0 & 0.2165 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.7}
$$

c)

Use the homogeneous transformation matrix in your answer to (b) above to compute the angle between y axis of the base frame and that in the 3rd joint frame.
Knowing

$$
{}^{i}\boldsymbol{R}_j = \begin{bmatrix} \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{x}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{x}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{x}}_i \\ \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{y}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{y}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{y}}_i \\ \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{z}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{z}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{z}}_i \end{bmatrix} \tag{3.8}
$$

The orientation of the last link relative to the base frame is given by $c^{-1}(-0.3536) = 110.78°$

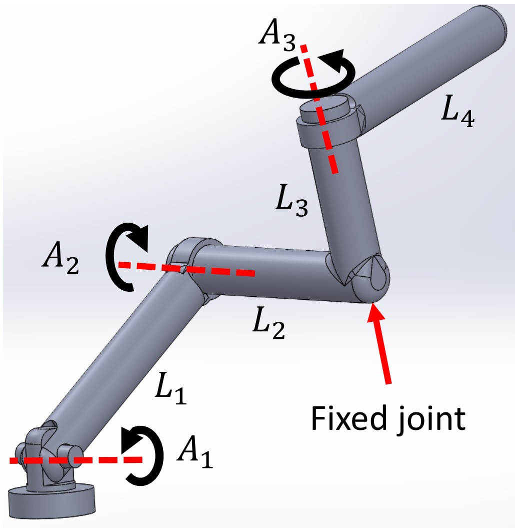


Figure 3.3: A 3-link robot manipulator with three revolute joints. The motors rotate around A_1-A_3 axises. The three links are L_1, L_2, L_3 and L_4 long respectively.

3.2 2017 past paper

Q2: Angular and end-point speeds

This question will let you determine positions and orientations of a 3-link manipulator.

a)

Use Denavit-Hartenberg (DH) convention to assign frames to each joint and fill up the DH table.

b)

Use the DH table in your answer to (a) above and the homogeneous transformation matrix in equation 3.10 to compute the position of the tip of the manipulator when $L_1 = 0.5$m, $L_2 = 0.25$m, $L_3 = 0.25$m, $L_4 = 0.5$m $\theta_1 = \pi/4$ rad, $\theta_2 = \pi/3$ rad, and $\theta_3 = \pi/3$ rad. Please note: θ_i, $i = 1, 2, 3$ are joint angles, d_i is the distance from $\hat{\boldsymbol{x}}_{i-1}$ to $\hat{\boldsymbol{x}}_i$ measured along $\hat{\boldsymbol{z}}_i$, α_i is the angle between $\hat{\boldsymbol{z}}_{i-1}$ and $\hat{\boldsymbol{z}}_i$ measured about $\hat{\boldsymbol{x}}_{i-1}$, and a_i is the distance from $\hat{\boldsymbol{z}}_{i-1}$ to $\hat{\boldsymbol{z}}_i$ measured along $\hat{\boldsymbol{x}}_{i-1}$.

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.9}
$$

c)

Use the homogeneous transformation matrix in your answer to (b) above to compute the angle between z axis of the base frame and that in the 3rd joint frame.

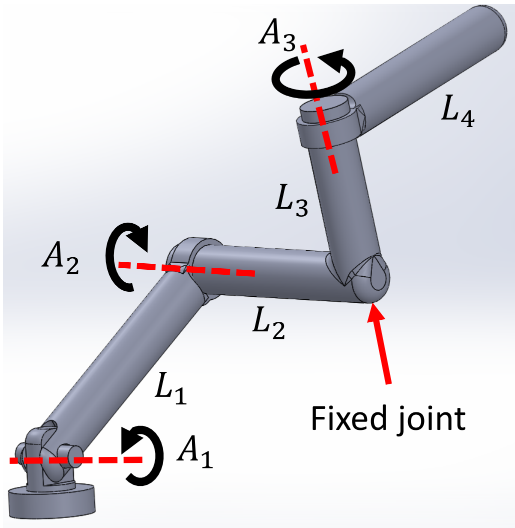


Figure 3.4: A robot manipulator with three revolute joints. The motors rotate around A_1-A_3 axises. There are four links of lengths L_1, L_2, L_3 and L_4.

Answers:

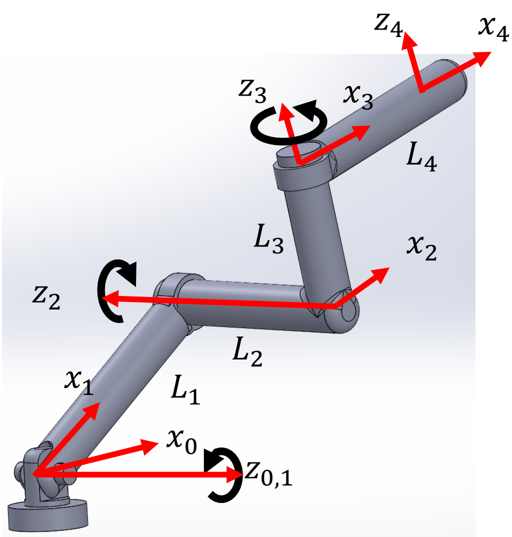
Q2: End-point position and orientation

This question will let you determine positions and orientations of a 3-link manipulator.

a)

Use Denavit-Hartenberg (DH) convention to assign frames to each joint and fill up the DH table.
[**30 marks**]

Answer:



i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	L_1	-180	$-L_2$	θ_2
3	0	90	L_3	θ_3
4	L_4	0	0	0

b)

Use the DH table in your answer to (a) above and the homogeneous transformation matrix in equation 3.10 to compute the position of the tip of the manipulator when $L_1 = 0.5m$, $L_2 = 0.25m$, $L_3 = 0.25m$, $L_4 = 0.5m$ $\theta_1 = \pi/4$ rad, $\theta_2 = \pi/3$ rad, and $\theta_3 = \pi/3$ rad. Please note: $\theta_i,\ \ i = 1, 2, 3$ are joint angles, d_i is the distance from $\hat{\boldsymbol{x}}_{i-1}$ to $\hat{\boldsymbol{x}}_i$ measured along $\hat{\boldsymbol{z}}_i$, α_i is the angle between $\hat{\boldsymbol{z}}_{i-1}$ and $\hat{\boldsymbol{z}}_i$ measured about $\hat{\boldsymbol{x}}_{i-1}$, and a_i is the distance from $\hat{\boldsymbol{z}}_{i-1}$ to $\hat{\boldsymbol{z}}_i$ measured along $\hat{\boldsymbol{x}}_{i-1}$.

$$ {}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.10} $$

[**50 marks**]
Answer:

$$ {}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.11} $$

$$ {}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & L_1 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & -1 & L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.12} $$

$$ {}^{2}\boldsymbol{T}_3 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ 0 & 0 & -1 & -L_3 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.13} $$

$$ {}^{3}\boldsymbol{T}_4 = \begin{bmatrix} 1 & 0 & 0 & L_4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.14} $$

${}^{0}\boldsymbol{T}_4 =^{0} \boldsymbol{T}_1\ {}^{1}\boldsymbol{T}_2\ {}^{2}\boldsymbol{T}_3\ {}^{3}\boldsymbol{T}_4$

$$ {}^{0}\boldsymbol{T}_4 = \begin{bmatrix} \cos(\theta_1-\theta_2)\cos(\theta_3) & -\cos(\theta_1-\theta_2)\sin(\theta_3) & -\sin(\theta_1-\theta_2) & L1\cos(\theta_1) - L3\sin(\theta_1-\theta_2) + L4\cos(\theta_1-\theta_2)\cos(\theta_3) \\ \sin(\theta_1-\theta_2)\cos(\theta_3) & -\sin(\theta_1-\theta_2)\sin(\theta_3) & \cos(\theta_1-\theta_2) & L1\sin(\theta_1) + L3\cos(\theta_1-\theta_2) + L4\sin(\theta_1-\theta_2)\cos(\theta_3) \\ -sin(\theta_3) & -cos(\theta_3) & 0 & L2 - L4\sin(\theta_3) \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.15} $$

$$ {}^{0}\boldsymbol{T}_4 = \begin{bmatrix} 0.4829 & -0.8365 & 0.2588 & 0.6597 \\ -0.1294 & 0.2241 & 0.9659 & 0.5303 \\ -0.8660 & -0.5 & 0 & -0.1830 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.16} $$

c)

Use the homogeneous transformation matrix in your answer to (b) above to compute the angle between x axis of the base frame and z in the 3rd joint frame. [**20 marks**]

Answer: Knowing the rotation from jth frame to ith frame given by,

$$^{i}\boldsymbol{R}_j = \begin{bmatrix} \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{x}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{x}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{x}}_i \\ \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{y}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{y}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{y}}_i \\ \hat{\boldsymbol{x}}_j \cdot \hat{\boldsymbol{z}}_i & \hat{\boldsymbol{y}}_j \cdot \hat{\boldsymbol{z}}_i & \hat{\boldsymbol{z}}_j \cdot \hat{\boldsymbol{z}}_i \end{bmatrix} \tag{3.17}$$

From equation 3.15 we can notice that the projection between the x axis of the base frame and z in the 3rd joint frame is given by the (1,3) term: $-\sin(\theta_1 - \theta_2) = \cos(90 + (\theta_1 - \theta_2))$, which means the angle is $90 + (\theta_1 - \theta_2)$. This can also be verified by visual inspection.

3.3 2018 past paper

Q2: 3D coordinate transformation for a mobile robot manipulator

This question will let you determine positions and orientations of a 2-link mobile manipulator.

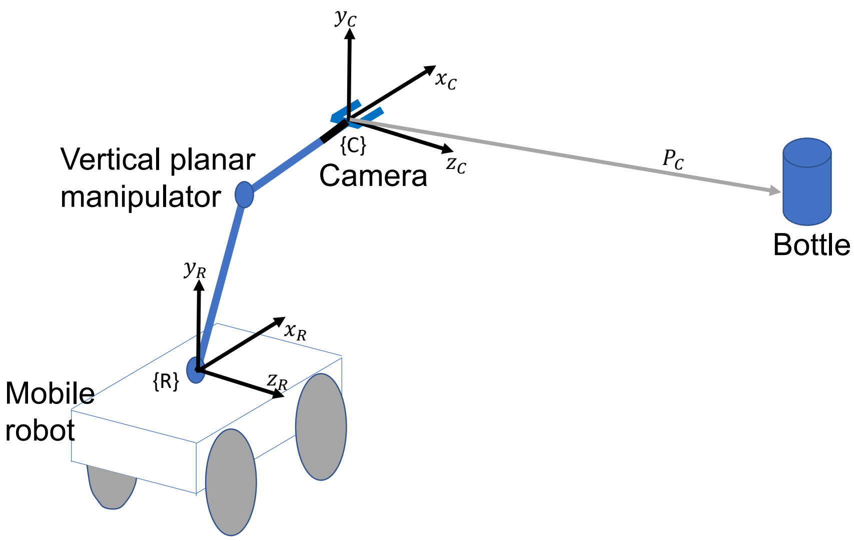


Figure 3.5: A 2-link planar robot manipulator mounted on a mobile robot. The two links are L_1 and L_2 long. The manipulator can move only in the vertical plane. A camera is mounted at the gripper of the manipulator.

a)

Use Denavit-Hartenberg (DH) convention to assign frames to the 2-link manipulator in figure 3.5 and fill up the DH table. [**20 marks**]

b)

Use the DH table in your answer to (Q2 (a)) above and the homogeneous transformation matrix from frame i to frame $i-1$ in equation 3.18 to obtain the transformation matrix from camera coordinates {C} to the mobile robot base coordinate system {R}. [**30 marks**]

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.18}
$$

c)

If the camera at the gripper of the 2-link manipulator sees a vertical bottle at $[0.5 \quad -0.6 \quad -1]^T$, when $\theta_1 = 60°$, $\theta_2 = 30°$, $L_1 = 0.75$m, and $L_2 = 0.5$m, calculate the vector coordinates of the bottle in robot base coordinate system {R}. Note: $\cos(30) = 0.866$ and $\cos(60) = 0.5$. [**30 marks**]

d)

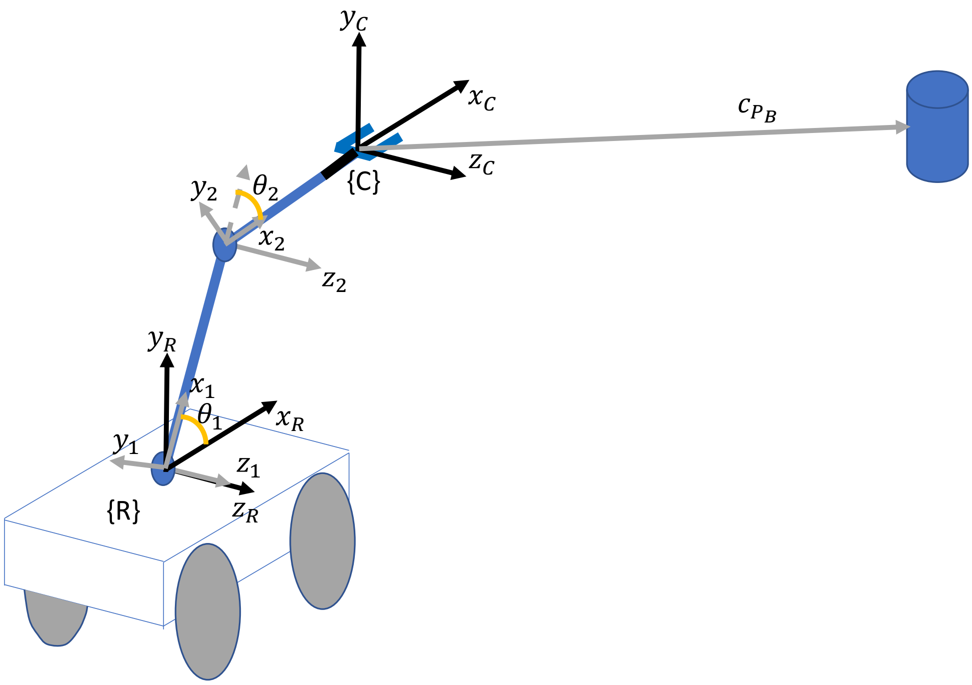
Use your result in Q2(c) to compute the angle the mobile robot should rotate to bring the 2-link manipulator and the bottle in the same plane. [**20 marks**]

Answers:

a)

Use Denavit-Hartenberg (DH) convention to assign frames to the 2-link manipulator in figure 3.5 and fill up the DH table. [**20 marks**]

Answer:



i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	L_1	0	0	θ_2
3	L_2	0	0	0

b)

Use the DH table in your answer to (Q2 (a)) above and the homogeneous transformation matrix from frame i to frame $i-1$ in equation 3.29 to obtain the transformation matrix from camera coordinates $\{C\}$ to the mobile robot base coordinate system $\{R\}$.

$$^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.19}$$

[**30 marks**]

Answer:

$$^{R}\boldsymbol{T}_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.20}$$

$$
{}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & L_1 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.21}
$$

$$
{}^{2}\boldsymbol{T}_C = \begin{bmatrix} 1 & 0 & 0 & L_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.22}
$$

${}^{R}\boldsymbol{T}_C ={}^{R} \boldsymbol{T}_1\ {}^{1}\boldsymbol{T}_2\ {}^{2}\boldsymbol{T}_C$

$$
{}^{R}\boldsymbol{T}_C = \begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & 0 & L1\cos(\theta_1)+L2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & 0 & L1\sin(\theta_1)+L2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.23}
$$

c)

If the camera at the gripper of the 2-link manipulator sees a vertical bottle at $[0.5 \quad -0.6 \quad -1]^T$*, when* $\theta_1 = 60°$*,* $\theta_2 = 30°$*,* $L_1 = 0.75m$*, and* $L_2 = 0.5m$*, calculate the vector coordinates of the bottle in robot base coordinate system* $\{R\}$*. Note:* $\cos(30) = 0.866$ *and* $\cos(60) = 0.5$*.* [**30 marks**]

Answer: ${}^{R}\boldsymbol{T}_B ={}^{R} \boldsymbol{T}_C\ \ {}^{C}P_B$, where ${}^{R}\boldsymbol{T}_B$ is the matrix transforming bottle frame {B} to mobile robot base frame {R}, ${}^{R}\boldsymbol{T}_C$ is that transforming camera frame {C} to mobile robot base frame {R}, and ${}^{C}P_B$ is the position of the bottle in camera coordinates.

Then, from equation (3.33), we obtain the position of the bottle in mobile robot base frame given by,

$$
{}^{R}\boldsymbol{P}_B = \begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & 0 & L1\cos(\theta_1)+L2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & 0 & L1\sin(\theta_1)+L2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ -0.6 \\ -1 \\ 1 \end{bmatrix} \tag{3.24}
$$

$$
{}^{R}\boldsymbol{P}_B = \begin{bmatrix} 0.5\cos(\theta_1+\theta_2)+0.6\sin(\theta_1+\theta_2)+L1\cos(\theta_1)+L2\cos(\theta_1+\theta_2) \\ 0.5\sin(\theta_1+\theta_2)-0.6\cos(\theta_1+\theta_2)+L1\sin(\theta_1)+L2\sin(\theta_1+\theta_2) \\ -1 \\ 1 \end{bmatrix} \tag{3.25}
$$

When $\theta_1 = 60°$ and $\theta_2 = 30°$ this becomes,

$$
\begin{aligned}
{}^{R}\boldsymbol{P}_B &= \begin{bmatrix} 0+0.6+0.75*0.5+0.5*0 \\ 0.5-0+0.75*0.866+0.5 \\ -1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 0.975 \\ 1.6495 \\ -1 \\ 1 \end{bmatrix} \qquad (3.26)
\end{aligned}
$$

d)

Use your result in Q2(c) to compute the angle the mobile robot should rotate to bring the 2-link manipulator and the bottle in the same plane. [**20 marks**]

Answer: From equation 3.26 we can notice that mobile robot should turn by an angle given by $\arctan(-1/0.975) = -45.7^\circ$ to bring the 2-link manipulator and the bottle in the same plane.

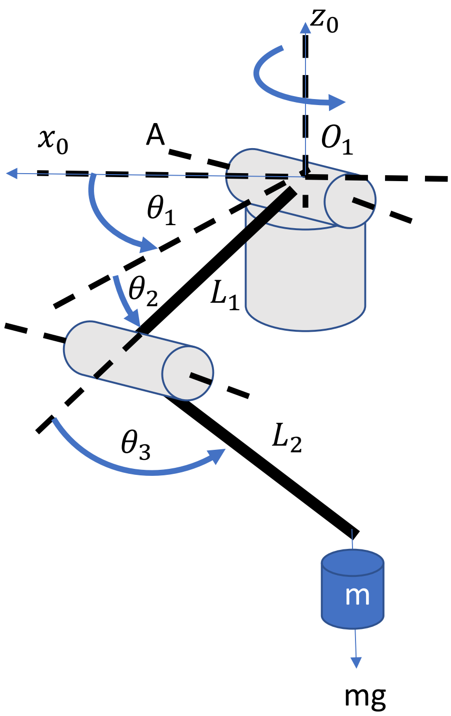


Figure 3.6: A robot manipulator driven by 3-motors. The first motor at the base has a vertical axis denoted by z_0, and its angle is θ_1. The second and third motors rotate around axes A and B respectively. Axis A is parallel to axis B. When the first motor angle $\theta_1 = 0$, the axis A of the second motor is perpendicular to axis x_0. A mass m is vertically hung from a string attached to the tip of the robot.

3.4 2019 paper: 3D coordinate transformation for a mobile robot manipulator

This question will let you determine positions and orientations of a 3 degrees of freedom manipulator and how its external forces are mapped to internal torques.

a)

Use Denavit-Hartenberg (DH) convention to assign frames to the robot manipulator in figure 3.7 and fill up the DH table. [**20 marks**]

b)

Use the DH table in your answer to question Q2(a) above and the homogeneous transformation matrix from frame i to frame $i-1$ in equation 3.56 to obtain the transformation matrix from the tip of the robot manipulator to the frame attached to the motor with axis A. [**30 marks**]

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.27}
$$

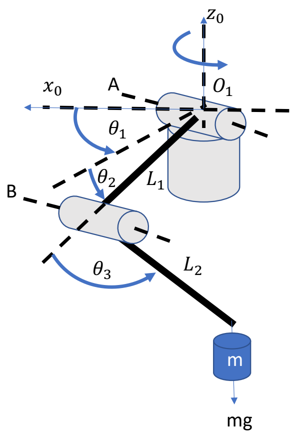


Figure 3.7: A robot manipulator driven by 3-motors. The first motor at the base has a vertical axis denoted by z_0, and its angle is θ_1. The second and third motors rotate around axes A and B respectively. Axis A is parallel to axis B. When the first motor angle $\theta_1 = 0$, the axis A of the second motor is perpendicular to axis x_0. A mass m is vertically hung from a string attached to the tip of the robot.

c)

The external force vector at the tip of the robot is given by $\boldsymbol{f} = [0\ 0\ -mg]^T$ where, m is the mass hanging at the tip, and $g = 9.8$ ms^{-2}. The resulting 2 joint torques are given by $\boldsymbol{\tau} = [\tau_A \ \ \tau_B]^T$, where τ_A and τ_B are torques around axes A and B. The Jacobian matrix that relates the external force to joint torques according to $\boldsymbol{\tau}_{2\times1} = \left[\boldsymbol{J}^T\right]_{2\times3} \boldsymbol{f}_{3\times1}$ is given by

$$\boldsymbol{J} = \begin{bmatrix} -L_1\sin(\theta_2) - L_2\sin(\theta_2+\theta_3) & -L_2\sin(\theta_2+\theta_3) \\ 0 & 0 \\ -L_1\cos(\theta_2) - L_2\cos(\theta_2+\theta_3) & -L_2\cos(\theta_2+\theta_3) \end{bmatrix} \tag{3.28}$$

Use equation 3.34 to derive an equation for the torque around axis B. Hence show a joint configuration where the torque around axis B is zero.
[**20 marks**]

d)

Show how the speed of the base motor $\dot{\theta}_1$ will affect the torque around axis B when $\theta_2 = 0$. Assume the horizontal distance from the vertical axis z_0 to the mass m is $L_2\cos(\theta_3) + L_1$ when $\theta_2 = 0$.

Hint: When a mass m is rotated at $\dot{\theta}$ with a radius r, the radial outward force (centrifugal force) on the mass is given by $mr\dot{\theta}^2$. [**30 marks**]

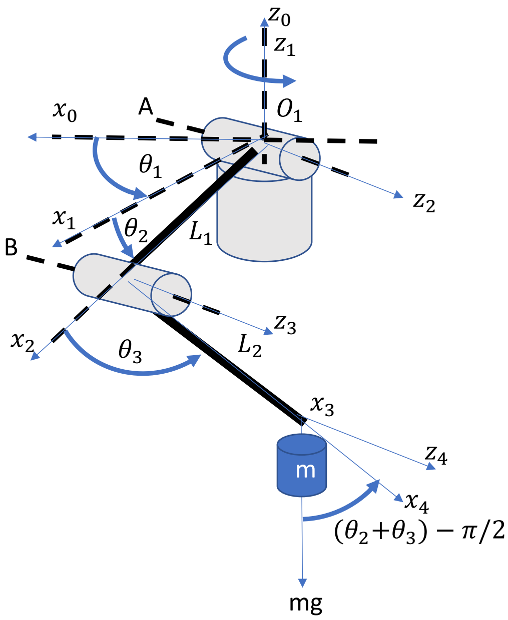
Answers to Robotics Q2 2019

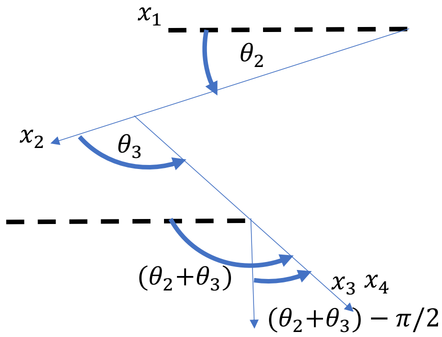
This question will let you determine positions and orientations of a 3 degrees of freedom manipulator and how its external forces are mapped to internal torques.

a)

*Use Denavit-Hartenberg (DH) convention to assign frames to the robot manipulator in figure 3.7 and fill up the DH table. [**20 marks**]*

Answer:





i	a_i	α_i	d_i	θ_i
1	0	0	0	θ_1
2	0	$-\pi/2$	0	θ_2
3	L_1	0	0	θ_3
4	L_2	0	0	0

b)

Use the DH table in your answer to (Q2 (a)) above and the homogeneous transformation matrix from frame i to frame $i-1$ in equation 3.29 to obtain the transformation matrix from the tip of the robot manipulator to the frame attached to the motor with axis A. [**30 marks**]

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.29}
$$

[**20 marks**]

Answer:

$$
{}^{3}\boldsymbol{T}_4 = \begin{bmatrix} 1 & 0 & 0 & L_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.30}
$$

$$
{}^{2}\boldsymbol{T}_3 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & L_1 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.31}
$$

$$
{}^1\boldsymbol{T}_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.32}
$$

${}^1\boldsymbol{T}_4 = {}^1\boldsymbol{T}_2\ {}^2\boldsymbol{T}_3\ {}^3\boldsymbol{T}_4$

$$
{}^1\boldsymbol{T}_4 = \begin{bmatrix} \cos(\theta_2+\theta_3) & -\sin(\theta_2+\theta_3) & 0 & L_2\cos(\theta_2+\theta_3) + L_1\cos(\theta_2) \\ 0 & 0 & 1 & 0 \\ -\sin(\theta_2+\theta_3) & -\cos(\theta_2+\theta_3) & 0 & -L_2\sin(\theta_2+\theta_3) - L_1\sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.33}
$$

c)

The external force vector at the tip of the robot is given by $\boldsymbol{f} = [0\ 0\ \ -mg]^T$ where, m is the mass hanging at the tip, and $g = 9.8\ ms^{-2}$. The resulting 2 joint torques are given by $\boldsymbol{\tau} = [\tau_A\ \ \tau_B]^T$, where τ_A and τ_B are torques around axes A and B. The Jacobian matrix that relates the external force to joint torques according to $\boldsymbol{\tau}_{2\times1} = \left[\boldsymbol{J}^T\right]_{2\times3} \boldsymbol{f}_{3\times1}$ is given by

$$
\boldsymbol{J} = \begin{bmatrix} -L_1\sin(\theta_2) - L_2\sin(\theta_2+\theta_3) & -L_2\sin(\theta_2+\theta_3) \\ 0 & 0 \\ -L_1\cos(\theta_2) - L_2\cos(\theta_2+\theta_3) & -L_2\cos(\theta_2+\theta_3) \end{bmatrix} \tag{3.34}
$$

Use equation 3.34 to derive an equation for the torque around axis B. Hence show a joint configuration where the torque around axis B is zero.

*[**20 marks**]*

Answer: The force vector when the base motor does not rotate is given by

$$
\boldsymbol{f} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} \tag{3.35}
$$

Then, $\boldsymbol{\tau}_{2\times1} = \left[\boldsymbol{J}^T\right]_{2\times3} \boldsymbol{f}_{3\times1}$ gives,

$$
\begin{aligned}
\boldsymbol{\tau} &= \begin{bmatrix} -L_1\sin(\theta_2) - L_2\sin(\theta_2+\theta_3) & 0 & -L_1\cos(\theta_2) - L_2\cos(\theta_2+\theta_3) \\ -L_2\sin(\theta_2+\theta_3) & 0 & -L_2\cos(\theta_2+\theta_3) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} && (3.36) \\
&= \begin{bmatrix} mg\left(L_2\cos(\theta_2+\theta_3) + L_1\cos(\theta_2)\right) \\ mg\left(L_2\cos(\theta_2+\theta_3)\right) \end{bmatrix} && (3.37)
\end{aligned}
$$

Therefore, the torque around axis B, $\tau_B = mg\left(L_2\cos(\theta_2+\theta_3)\right)$.

We can see that $\tau_B = 0$ when $(\theta_2+\theta_3) = 90°$. Physically, this can happen when $\theta_2 = 0$ and $\theta_3 = 90°$.

d)

Show how the speed of the base motor $\dot{\theta}_1$ will affect the torque around axis B when $\theta_2 = 0$. Assume the horizontal distance from the vertical axis z_0 to the mass m is $L_2\cos(\theta_3) + L_1$ when $\theta_2 = 0$.

*Hint: When a mass m is rotated at $\dot{\theta}$ with a radius r, the radial outward force (centrifugal force) on the mass is given by $mr\dot{\theta}^2$. [**30 marks**]*

Answer: The force vector when the base motor rotates at $\dot{\theta}_1$ with $\ddot{\theta}_1 = 0$ is given by

$$\boldsymbol{f} = \begin{bmatrix} m(L_2\cos(\theta_3) + L_1)\dot{\theta}_1^2 \\ 0 \\ -mg \end{bmatrix} \tag{3.38}$$

Then the torque vector is given by

$$\begin{aligned}\boldsymbol{\tau} &= \begin{bmatrix} -L_2\sin(\theta_3) & 0 & -L_1 - L_2\cos(\theta_3) \\ -L_2\sin(\theta_3) & 0 & -L_2\cos(\theta_3) \end{bmatrix} \begin{bmatrix} m(L_2\cos(\theta_3) + L_1)\dot{\theta}_1^2 \\ 0 \\ -mg \end{bmatrix} & (3.39) \\ &= \begin{bmatrix} mg\left(L_2\cos(\theta_3) + L_1\right) - L_2\left(m(L_2\cos(\theta_3) + L_1)\dot{\theta}_1^2\right)\sin(\theta_3) \\ mgL_2\cos(\theta_3) - L_2\left(m(L_2\cos(\theta_3) + L_1)\dot{\theta}_1^2\right)\sin(\theta_3) \end{bmatrix} & (3.40)\end{aligned}$$

Therefore, we can see from equation 3.40 that τ_B has changed from
$\tau_B = mg\left(L_2\cos(\theta_2 + \theta_3)\right)$ to $\tau_B = mgL_2\cos(\theta_3) - L_2\left(m(L_2\cos(\theta_3) + L_1)\dot{\theta}_1^2\right)\sin(\theta_3)$
which has a term $L_2\left(m(L_2\cos(\theta_3) + L_1)\dot{\theta}_1^2\right)\sin(\theta_3)$ affecting the value of τ_B due to the rotation of the base axis. For instance, when $\theta_3 < 90°$ we can be sure that this term will reduce the torque at axis B when the base rotates.

3.5 2020 paper: 3D coordinate transformation for soccer robots

This question will let you use quaternions to plan movements of a soccer robot.

a)

Given a unit quaternion

$$\boldsymbol{q} = \epsilon_1 i + \epsilon_2 j + \epsilon_3 k + \epsilon_4$$

such that $\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 = 1$, the rotation matrix that rotates vector v in frame A (denoted by ${}^A v$) to that in frame B (denoted by ${}^B v$) is given by

$$\begin{aligned}{}^B v &= \boldsymbol{q}\ {}^A v\ \boldsymbol{q}^* \\ &= R_\epsilon\ {}^A v \end{aligned} \tag{3.41}$$

Then the rotation matrix R_ϵ can be written as

$$R_\epsilon = \begin{bmatrix} 1 - 2\epsilon_2^2 - 2\epsilon_3^2 & 2\left(\epsilon_1\epsilon_2 - \epsilon_3\epsilon_4\right) & 2\left(\epsilon_1\epsilon_3 + \epsilon_2\epsilon_4\right) \\ 2\left(\epsilon_1\epsilon_2 + \epsilon_3\epsilon_4\right) & 1 - 2\epsilon_1^2 - 2\epsilon_3^2 & 2\left(\epsilon_2\epsilon_3 - \epsilon_1\epsilon_4\right) \\ 2\left(\epsilon_1\epsilon_3 - \epsilon_2\epsilon_4\right) & 2\left(\epsilon_2\epsilon_3 + \epsilon_1\epsilon_4\right) & 1 - 2\epsilon_1^2 - 2\epsilon_2^2 \end{bmatrix}$$

If the unit quaternion of rotating the orientation of camera C_R to that of camera C_G is given by

$$\boldsymbol{q} = 0.3062i + 0.3062j + 0.25k + 0.866 \tag{3.42}$$

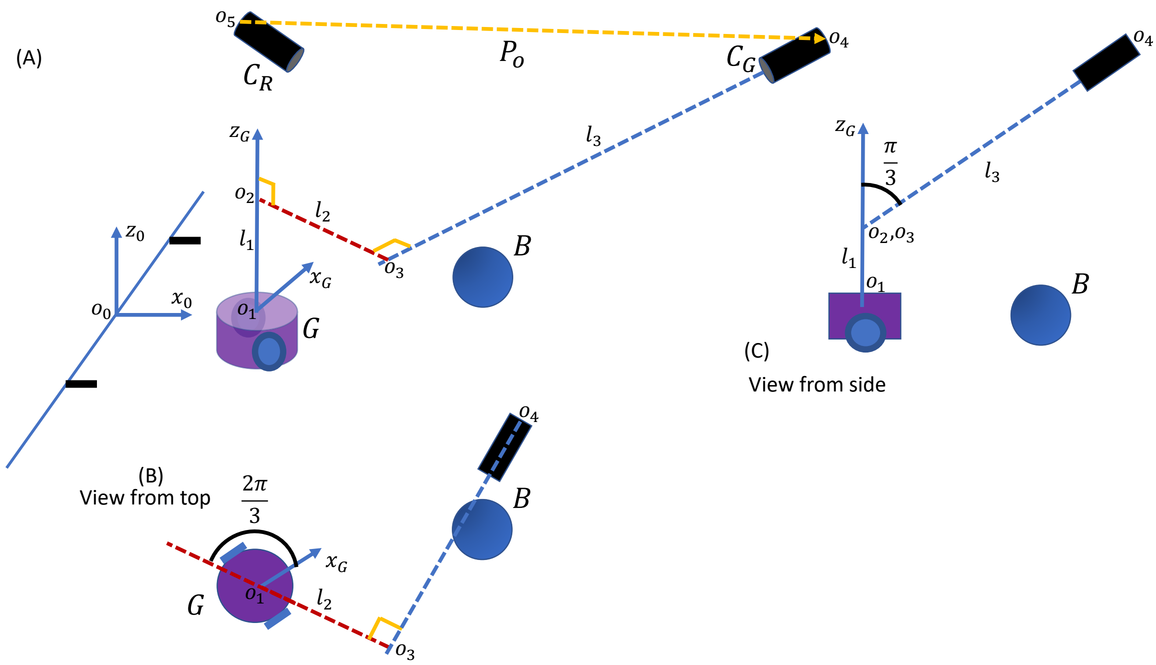


Figure 3.8: Two cameras C_R and C_G keep tracking soccer robot G. The aim of competing robots (not shown) is to hit the ball B to reach the goal avoiding G. The robot G should move in the way of the ball to prevent it from reaching the goal. The common perpendicular between the vertical axis of robot G and the axis of camera C_G is shown in o_2o_3 line. The lengths of o_1o_2, o_2o_3, and o_3o_4 are l_1, l_2, and l_3 respectively.

such that the coefficient vector $\boldsymbol{q}_C = [0.3062,\ 0.3062,\ 0.25,\ 0.866]^T$ gives the following:

$$\boldsymbol{q}_C^T\boldsymbol{q}_C = 1 \tag{3.43}$$

and

$$\boldsymbol{q}_C\boldsymbol{q}_C^T = \begin{bmatrix} 0.0937 & 0.0937 & 0.0765 & 0.2652 \\ 0.0937 & 0.0937 & 0.0765 & 0.2652 \\ 0.0765 & 0.0765 & 0.0625 & 0.2165 \\ 0.2652 & 0.2652 & 0.2165 & 0.7500 \end{bmatrix} \tag{3.44}$$

If the origin of camera frame C_R in the C_G frame is given by

$$\boldsymbol{P}_o = \begin{bmatrix} 0.25 \\ 0 \\ -2 \end{bmatrix} \tag{3.45}$$

Find the homogeneous transformation matrix from frame C_R to C_G denoted by ${}^{C_G}T_{C_R}$.
[**20 marks**]

b)

Assign frames following the D-H convention to find the homogenous transformation matrix from frame C_G to G denoted by ${}^{G}T_{C_G}$. Then fill the D-H table using the following parameter values:

l_1	0.5
l_2	0.25
l_3	1.5
Angle between x_G and o_2o_3 around z_G	$2\frac{\pi}{3}$
Angle between z_G and o_3o_4 around o_2o_3	$\frac{\pi}{3}$

Hence find the homogenous transformation matrix from frame C_G to G denoted by ${}^{G}T_{C_G}$ following the format given in equation 3.56.

$$
{}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.46}
$$

[**30 marks**]

c)

Use ${}^{C_G}T_{C_R}$ from question Q2a and ${}^{G}T_{C_G}$ from Q2b to find the homogeneous vector of the ball B in frame G denoted by ${}^{G}v_B$ if the vector from the ball to the camera C_R denoted by ${}^{C_R}v_B$ is given by

$$
{}^{C_R}v_B = \begin{bmatrix} -2 \\ 0.5 \\ 1 \end{bmatrix} \tag{3.47}
$$

[**30 marks**]

d)

Mention two computational challenges in a realtime robot soccer team suggesting a measure you would recommend to face the respective challenges.

[**20 marks**]

Answers to Robotics Q2 2020

This question will let you use quaternions to plan movements of a soccer robot.

a)

Given a unit quaternion

$$
\boldsymbol{q} = \epsilon_1 i + \epsilon_2 j + \epsilon_3 k + \epsilon_4
$$

such that $\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 = 1$, the rotation matrix that rotates vector v in frame A (denoted by ${}^{A}v$) to that in frame B (denoted by ${}^{B}v$) is given by

$$
\begin{aligned}
{}^{B}v &= \boldsymbol{q}\ {}^{A}v\ \boldsymbol{q}^* \\
&= R_\epsilon\ {}^{A}v
\end{aligned} \tag{3.48}
$$

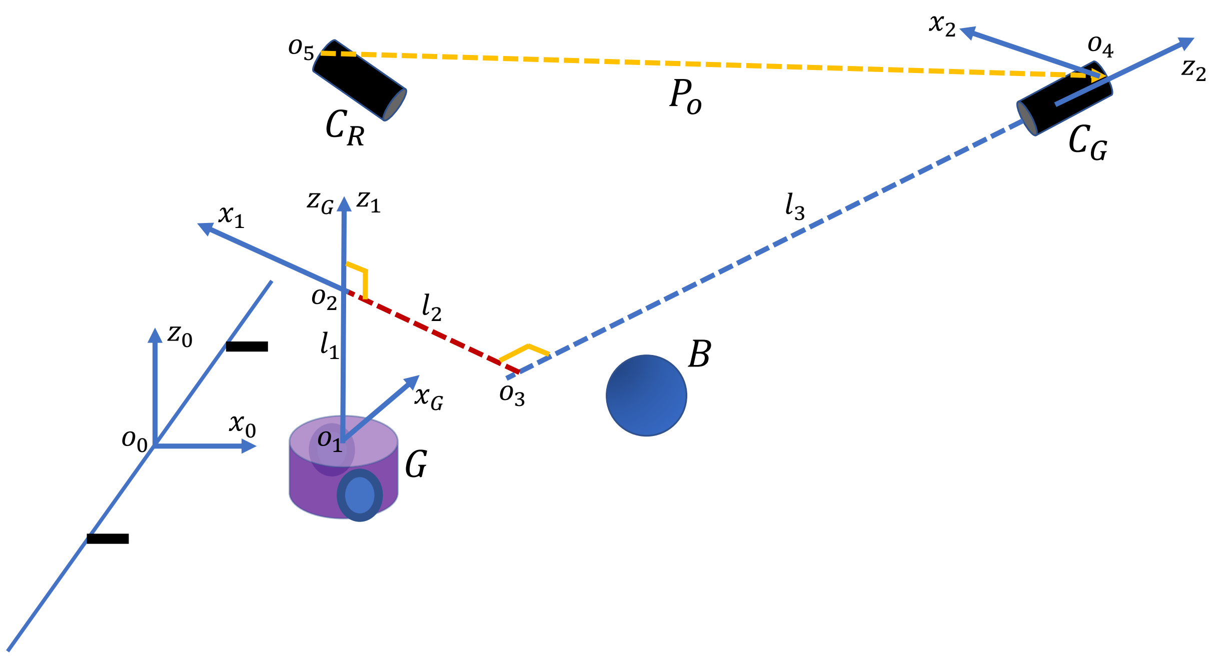


Figure 3.9: Two cameras C_R and C_G keep tracking soccer robot R and G. The aim or robot R is to hit the ball B to reach the goal avoiding G. The robot G should move in the way of the ball to prevent it from reaching the goal. The common perpendicular between the vertical axis of robot G and the axis of camera C_G is shown in o_2o_3 line. The lengths of o_1o_2, o_2o_3, and o_3o_4 are l_1, l_2, and l_3 respectively.

Then the rotation matrix R_ϵ can be written as

$$R_\epsilon = \begin{bmatrix} 1-2\epsilon_2^2-2\epsilon_3^2 & 2\left(\epsilon_1\epsilon_2-\epsilon_3\epsilon_4\right) & 2\left(\epsilon_1\epsilon_3+\epsilon_2\epsilon_4\right) \\ 2\left(\epsilon_1\epsilon_2+\epsilon_3\epsilon_4\right) & 1-2\epsilon_1^2-2\epsilon_3^2 & 2\left(\epsilon_2\epsilon_3-\epsilon_1\epsilon_4\right) \\ 2\left(\epsilon_1\epsilon_3-\epsilon_2\epsilon_4\right) & 2\left(\epsilon_2\epsilon_3+\epsilon_1\epsilon_4\right) & 1-2\epsilon_1^2-2\epsilon_2^2 \end{bmatrix}$$

If the unit quaternion of rotating the orientation of camera C_R to that of camera C_G is given by

$$\boldsymbol{q} = 0.3062i + 0.3062j + 0.25k + 0.866 \tag{3.49}$$

such that the coefficient vector $\boldsymbol{q}_C = [0.3062\ 0.3062\ 0.25\ 0.866]^T$ gives the following:

$$\boldsymbol{q}_C^T\boldsymbol{q}_C = 1 \tag{3.50}$$

and

$$\boldsymbol{q}_C\boldsymbol{q}_C^T = \begin{bmatrix} 0.0937 & 0.0937 & 0.0765 & 0.2652 \\ 0.0937 & 0.0937 & 0.0765 & 0.2652 \\ 0.0765 & 0.0765 & 0.0625 & 0.2165 \\ 0.2652 & 0.2652 & 0.2165 & 0.7500 \end{bmatrix} \tag{3.51}$$

We notice that the matrix in equation (3.51) takes the following format:

$$A = \begin{bmatrix} \epsilon_1^2 & \epsilon_1\epsilon_2 & \epsilon_1\epsilon_3 & \epsilon_1\epsilon_4 \\ \epsilon_2\epsilon_1 & \epsilon_2^2 & \epsilon_2\epsilon_3 & \epsilon_2\epsilon_4 \\ \epsilon_3\epsilon_1 & \epsilon_3\epsilon_2 & \epsilon_3^2 & \epsilon_3\epsilon_4 \\ \epsilon_4\epsilon_1 & \epsilon_4\epsilon_2 & \epsilon_4\epsilon_3 & \epsilon_4^2 \end{bmatrix}$$

By substituting above values in equation (3.49), we obtain

$$R_\epsilon = \begin{bmatrix} 0.6875 & -0.2455 & 0.6834 \\ 0.6205 & 0.6875 & -0.3772 \\ -0.3772 & 0.6834 & 0.6250 \end{bmatrix}$$

If the origin of camera frame C_R in the C_G frame is given by

$$\boldsymbol{P}_o = \begin{bmatrix} 0.25 \\ 0 \\ -2 \end{bmatrix} \tag{3.52}$$

The homogeneous transformation matrix from frame C_R to C_G denoted by ${}^{C_G}T_{C_R}$ is given by

$$\begin{aligned} {}^{C_G}T_{C_R} &= \begin{bmatrix} R_\epsilon & \boldsymbol{P}_o \\ \boldsymbol{O}_{1\times 3} & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.6875 & -0.2455 & 0.6834 & 0.25 \\ 0.6205 & 0.6875 & -0.3772 & 0 \\ -0.3772 & 0.6834 & 0.6250 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned} \tag{3.53}$$

[**20 marks**]

b)

Assign frames following the D-H convention to find the homogenous transformation matrix from frame C_G to G denoted by ${}^{G}T_{C_G}$. Then fill the D-H table using the following parameter values:

l_1	0.5
l_2	0.25
l_3	1.5
Angle between x_G and o_2o_3 around z_G	$2\frac{\pi}{3}$
Angle between z_G and o_3o_4 around o_2o_3	$\frac{\pi}{3}$

The resulting D-H table is given by

i	a_i	α_i	d_i	θ_i
1	0	0	l_1	$2\frac{\pi}{3}$
2	l_2	$\frac{\pi}{3}$	l_3	0

Hence find the homogenous transformation matrix from frame C_G to G denoted by ${}^{G}T_{C_G}$ following the format given in equation 3.56.

$$ {}^{i-1}\boldsymbol{T}_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_i \\ s\theta_i c\alpha_i & c\theta_i c\alpha_i & -s\alpha_i & -s\alpha_i d_i \\ s\theta_i s\alpha_i & c\theta_i s\alpha_i & c\alpha_i & c\alpha_i d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.54} $$

By substituting values from D-H table 3.5 in equation (3.56) we obtain,

$$ {}^{G}\boldsymbol{T}_1 = \begin{bmatrix} -0.5 & -0.866 & 0 & 0 \\ 0.866 & -0.5 & 0 & 0 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.55} $$

$$ {}^{1}\boldsymbol{T}_{C_G} = \begin{bmatrix} 1 & 0 & 0 & 0.25 \\ 0 & 0.5 & -0.866 & -0.866 * 1.5 \\ 0 & 0.866 & 0.5 & 0.5 * 1.5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.56} $$

$$ {}^{G}\boldsymbol{T}_{C_G} = \begin{bmatrix} -0.5 & -0.43 & 0.75 & 0.99 \\ 0.866 & -0.25 & 0.43 & 0.866 \\ 0 & 0.866 & 0.5 & 1.25 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.57} $$

[**30 marks**]

c)

Use ${}^{C_G}T_{C_R}$ from question Q2a and ${}^{G}T_{C_G}$ from Q2b to find the homogeneous vector of the ball B in frame G denoted by ${}^{G}v_B$ if the vector from the ball to the camera C_R denoted by ${}^{C_R}v_B$ is given by

$$^{C_R}v_B = \begin{bmatrix} -2 \\ 0.5 \\ 1 \end{bmatrix} \tag{3.58}$$

We use compound transformations as given below:

$$\begin{aligned} ^{G}v_B &= {^{G}T_{C_G}} * {^{C_G}T_{C_R}} * {^{C_R}v_B} \\ &= \begin{bmatrix} -0.5 & -0.43 & 0.75 & 0.99 \\ 0.866 & -0.25 & 0.43 & 0.866 \\ 0 & 0.866 & 0.5 & 1.25 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.6875 & -0.2455 & 0.6834 & 0.25 \\ 0.6205 & 0.6875 & -0.3772 & 0 \\ -0.3772 & 0.6834 & 0.6250 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 0.5 \\ 1 \\ 1 \end{bmatrix} && (3.59) \\ &= \begin{bmatrix} 1.62 \\ 0.57 \\ 0 \\ 1 \end{bmatrix} && (3.60) \end{aligned}$$

Rounded up answers are acceptable such as

$$^{G}v_B = \begin{bmatrix} 1.6 \\ 0.6 \\ 0 \\ 1 \end{bmatrix} \tag{3.61}$$

[**30 marks**]

d)

Mention two computational challenges in a realtime robot soccer team suggesting a measure you would recommend to face the respective challenges.

Two computational challenges:

- Estimation of DH parameters in matrix $^{G}T_{C_g}$ realtime using camera images is a challenge for realtime performance.
- Computation of compound transformations to obtain $^{G}v_B$ in every sampling interval is a challenge.

Things we can do to mitigate the challenges:

- Since robots have inertia, and they usually travel along straight lines, a history of DH parameters can be used in a smoothing algorithm to predict future values. A cost function of accumulated errors of prediction can be used in a moving window to make predictions and corrections in a model based predictive algorithm.
- Since the ball usually travels along a straight line unless a robot hits it, use a prediction method to compute $^{G}v_B$ using low frequency sampling of DH parameters in matrix $^{G}T_{C_g}$ using camera images. The sampling frequency can be changed depending on the relative distance between a robot and the ball. If the ball is sufficiently away from a robot, the history of its movement can be used to predict the next location.

[**20 marks**]

Bibliography

[1] Etienne Dombre and Wisama Khalil. *Modeling, performance analysis and control of robot manipulators*. Wiley Online Library, 2007.

[2] Richard S Hartenberg and Jacques Denavit. A kinematic notation for lower pair mechanisms based on matrices. 1955.

[3] Bryan A Jones and Ian D Walker. Kinematics for multisection continuum robots. *IEEE Transactions on Robotics*, 22(1):43–55, 2006.

[4] SM Hadi Sadati, Ali Shiva, Nicolas Herzig, Caleb D Rucker, Helmut Hauser, Ian D Walker, Christos Bergeles, Kaspar Althoefer, and Thrishantha Nanayakkara. Stiffness imaging with a continuum appendage: Real-time shape and tip force estimation from base load readings. *IEEE Robotics and Automation Letters*, 5(2):2824–2831, 2020.